# Introduction to Rule Based Modeling with BioNetGen

**Jim Faeder**

*MSCBIO 2025, Introduction to Bioinformatics Programming in Python*

September 25, 2025

**Download and documentation: bionetgen.org**

More detailed intro to syntax

Lecture slides and code

Lecture video – Part 1

Lecture – Part 2

# Reaction Networks Models can be used to describe a wide range of natural processes.

**Chemistry**

*Ozone formation in the upper atmosphere*

$$O_2 \xrightarrow[h\upsilon]{k_1} 2\,O$$

$$O_2 + O \underset{k_3}{\overset{k_2}{\leftrightarrow}} O_3$$

$$Cl + O_3 \xrightarrow{k_4} ClO + O_2$$

$$ClO + O_3 \xrightarrow{k_5} Cl + 2\,O_2$$

**Epidemiology**

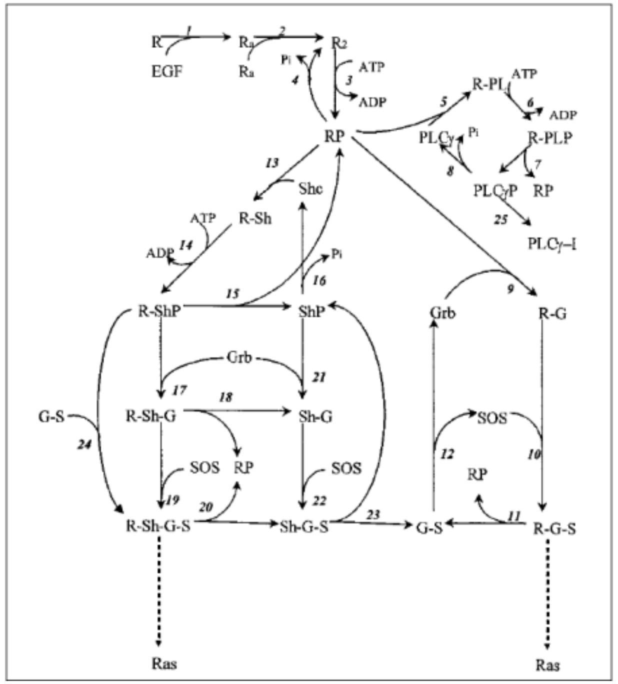
*Susceptible-Infected-Recovered (SIR)*

$$S + I \overset{\beta}{\rightarrow} I + I$$

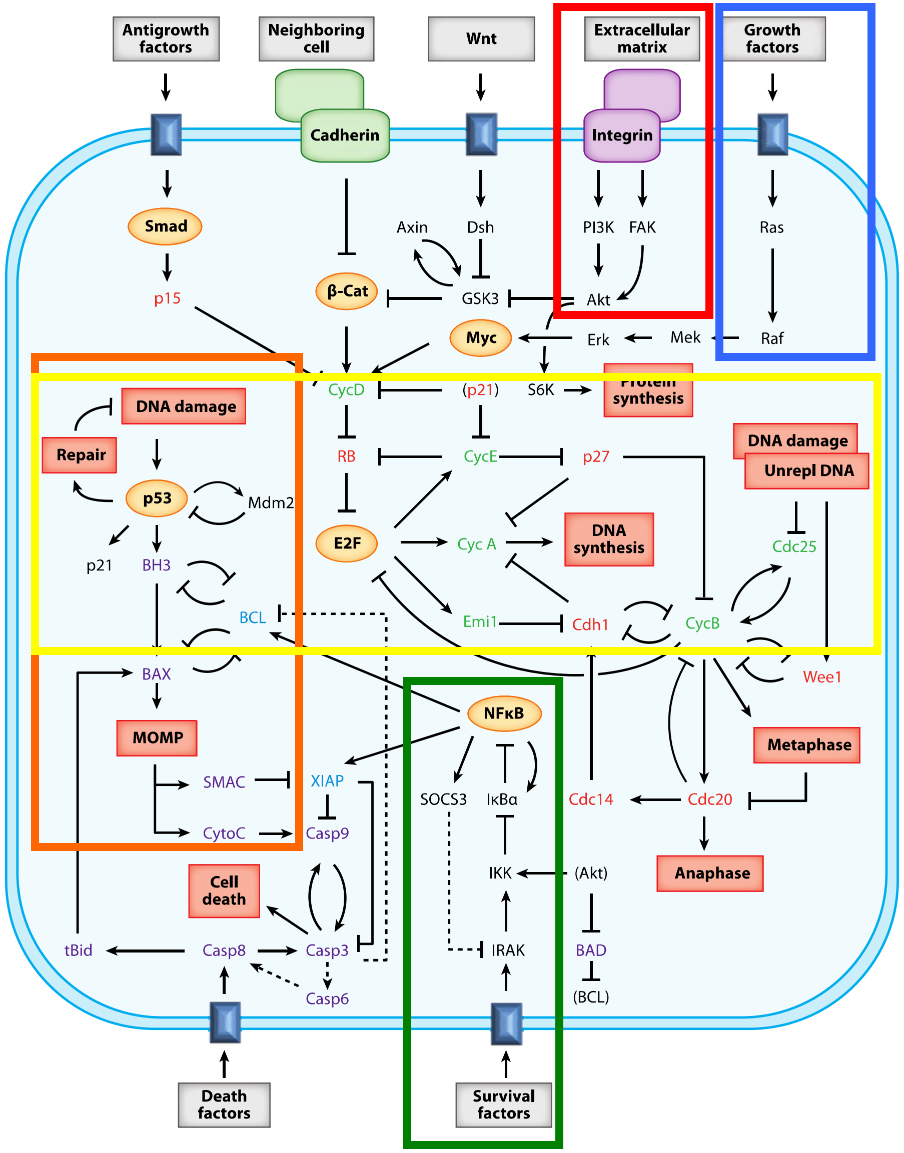
$$I \overset{\gamma}{\rightarrow} R$$

**Molecular biology**

*Signaling network*



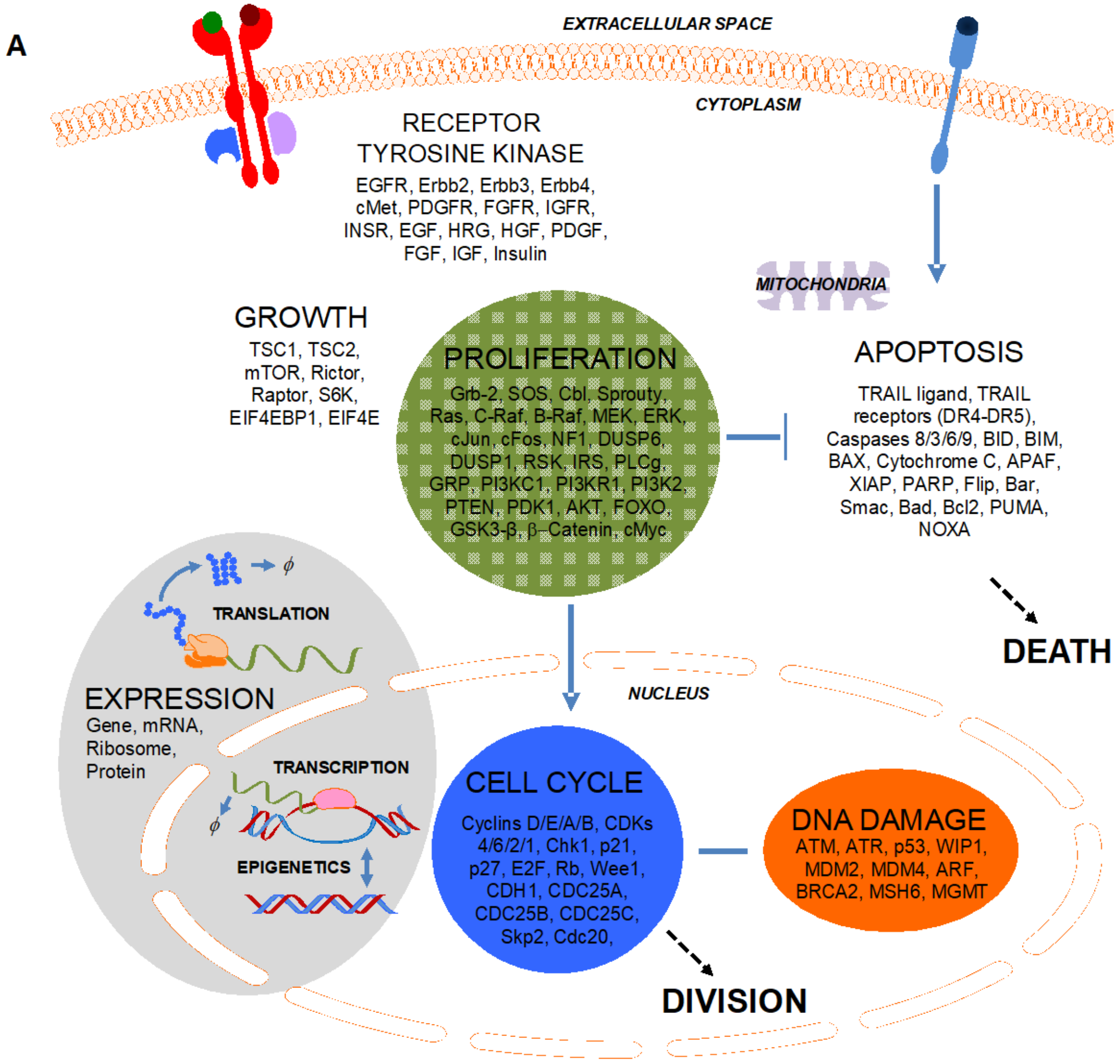
# Toward an integrated model of the cell

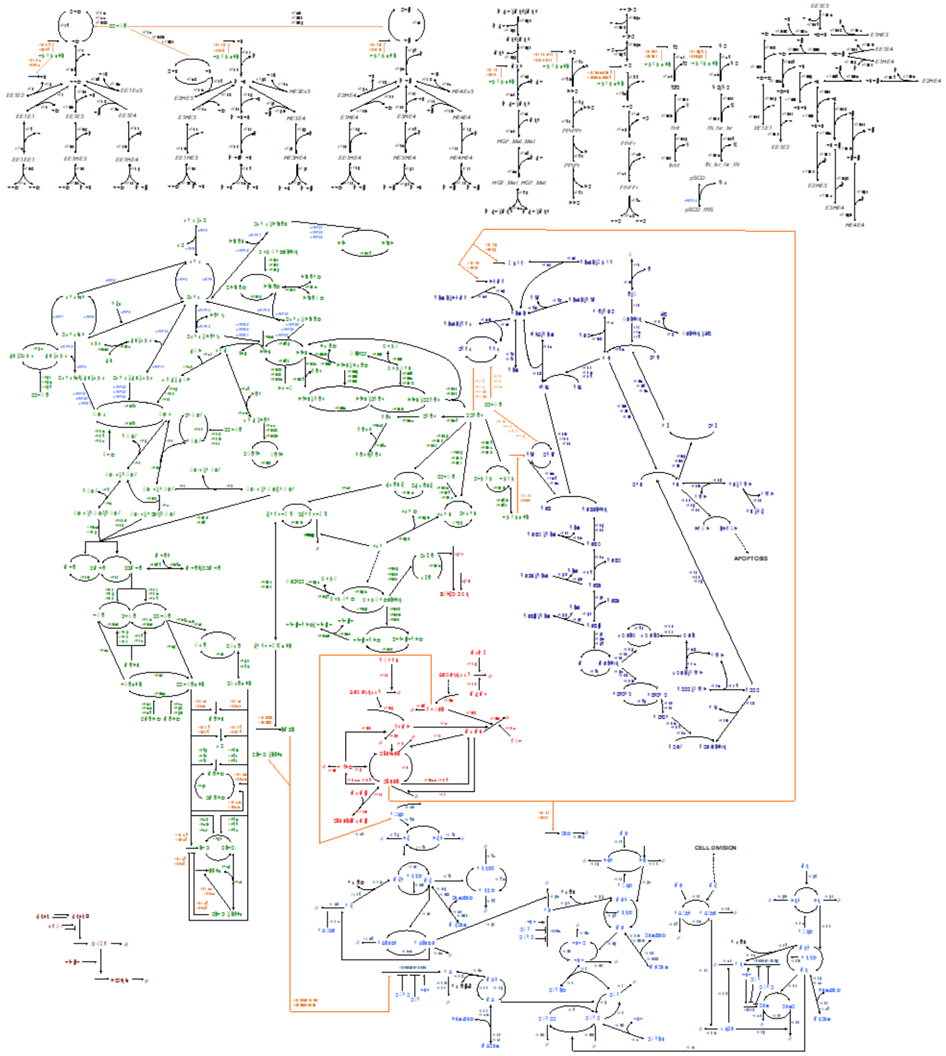



How to accomplish the task of linking together signaling modules that determine cell fate?

Tyson and Novak, *Annu. Rev. Phys. Chem.* **61**, 219–40 (2010)

# Integrated Model of MCF10A Cell



Bouhaddou et al. (2017) *PLOS Computational Biology*

# Susceptible-Infected-Recovered Model of Disease Spread

**Species**

*S* are "susceptible" individuals who may become infected
*I* are "infected" individuals who may infect *S*
*R* are "recovered" individuals who resist further infection

**Reactions**

*I* infect *S*: $S + I \xrightarrow{\beta} I + I$ (r1)

*I* recover: $I \xrightarrow{\gamma} R$ (r2)

The parameter $\beta$ is a **rate constant**, and the rate at which the reaction occurs is given by $r_1 = \beta \cdot S \cdot I$

$r_2 = \gamma \cdot I$

# Reaction network models (RNM) are built from individual reactions.

The basic SIR model contains two **reactions**

$$S + I \overset{\beta}{\rightarrow} I + I \qquad \text{(r1)}$$

$$I \overset{\gamma}{\rightarrow} R \qquad \text{(r2)}$$

and three **species**: $S$, $I$, and $R$, which represents the number of recovered individuals.

An RNM tracks the values of the species concentrations as a function of time, which can be written as the vector

$$\boldsymbol{X}(t) = \begin{pmatrix} S \\ I \\ R \end{pmatrix}$$

A complete RNM consists of a set of **reactions**, **species**, **rate constants**, and the **initial values** (also called **initial concentrations**) of the species, denoted $\boldsymbol{X}(0)$.

# Ordinary differential equations (ODEs) describe the mean behavior of a reaction network.

The SIR model contains two **reactions**

$$S + I \xrightarrow{\beta} I + I \qquad \text{(r1)}$$
$$I \xrightarrow{\gamma} R \qquad \text{(r2)}$$

which give rise to three ODE's

$$\begin{array}{rcc} & \text{r1} & \text{r2} \\ \frac{dS}{dt} = & -r_1 & \\ \frac{dI}{dt} = & +r_1 & -r_2 \\ \frac{dR}{dt} = & & +r_2 \end{array}$$

$$\boxed{\begin{aligned} \frac{dS}{dt} &= -\beta \cdot S \cdot I \\ \frac{dI}{dt} &= \beta \cdot S \cdot I - \gamma \cdot I \\ \frac{dR}{dt} &= \gamma \cdot I \end{aligned}}$$

where the rates are given by

$$r_1 = \beta \cdot S \cdot I$$
$$r_2 = \gamma \cdot I$$

# The ODEs for RNMs have a compact representation using the stoichiometry matrix.

$$\frac{d\boldsymbol{X}(t)}{dt} = \left[\begin{array}{rcc} & \text{r1} & \text{r2} \\ \frac{dS}{dt} = & \boxed{-r_1} & \\ \frac{dI}{dt} = & +r_1 & \boxed{-r_2} \\ \frac{dR}{dt} = & & +r_2 \end{array}\right] = \begin{pmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} r_1 \\ r_2 \end{pmatrix}$$

$$\mathbf{S} \quad \cdot \quad \mathbf{v}(\boldsymbol{X})$$

Stoichiometry matrix

Reaction fluxes

The elements of **S** are given by

$S_{ij} =$ (# times species $i$ appears as product in reaction $j$)
−(# times species $i$ appears as reactant in reaction $j$)

# ODEs for most RNMs are solved numerically.

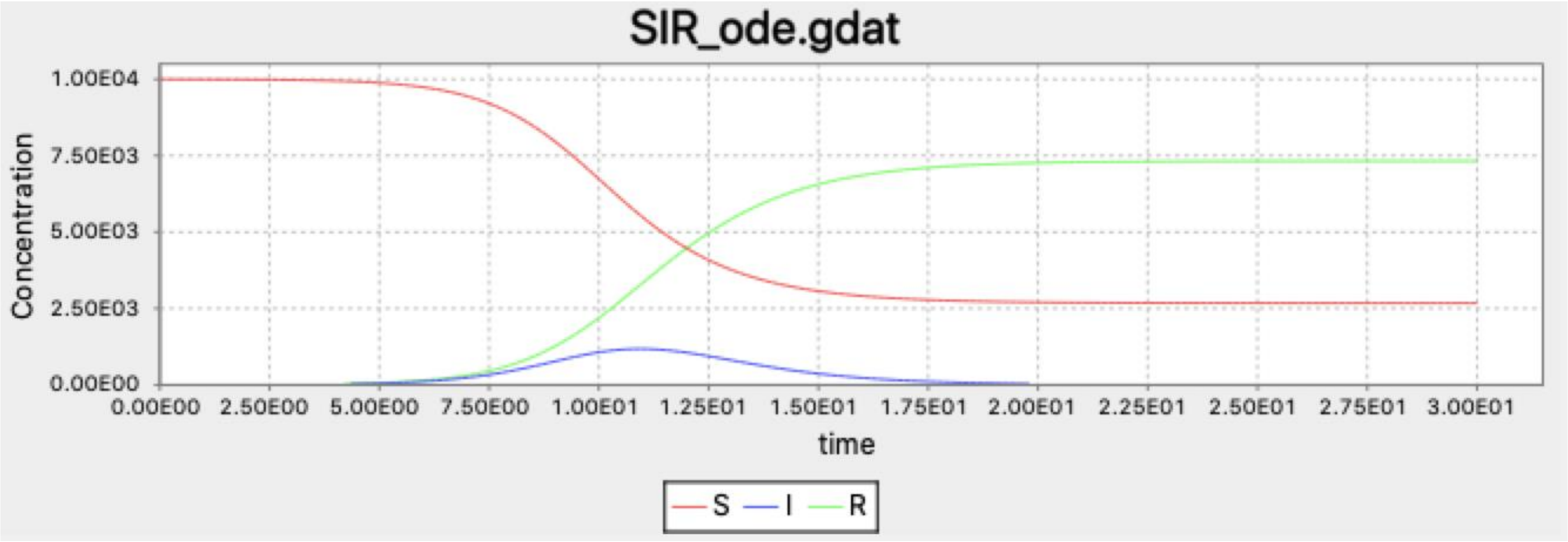
The general equation for RNM dynamics is

$$\frac{d\boldsymbol{X}(t)}{dt} = \mathbf{S}\cdot\mathbf{v}(\boldsymbol{X})$$

- These equations are in general *nonlinear.*
- They cannot be solved analytically except in special cases.
- Numerical integration is required to obtain a solution.
- “Stiffness” arises because reactions occur on different timescales, requiring special techniques to obtain accurate solutions efficiently.
- Widely used solvers are CVODE and LSODA packages.
- Accessed using ode method in the simulate command of BNG.

# Numerical solution of SIR model describes dynamics of infection outbreak.

Time course for *S*, *I*, and *R* produced by BioNetGen and displayed in RuleBender.



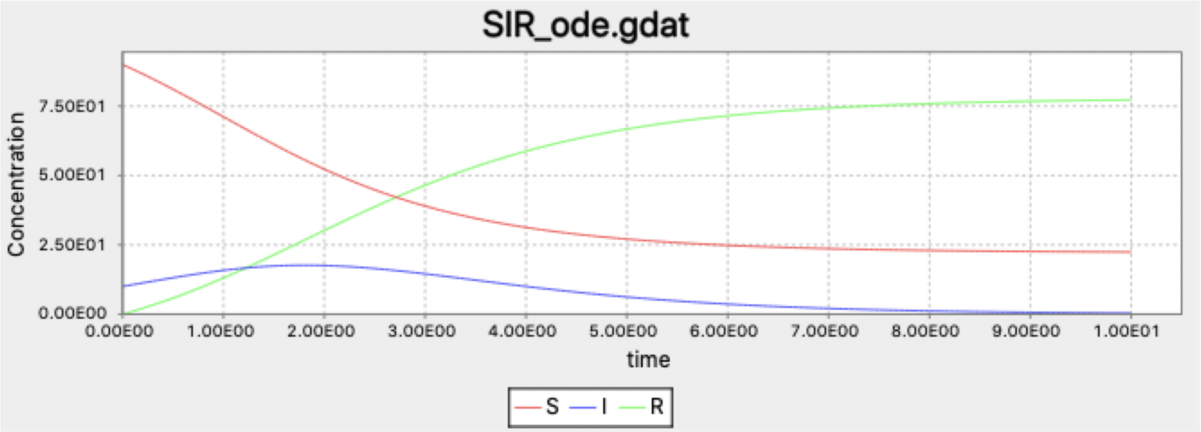

$$\begin{pmatrix}\beta\\ \gamma\end{pmatrix} = \begin{pmatrix}1.8\times 10^{-3}\\ 1\end{pmatrix}, \qquad \begin{pmatrix}S(0)\\ I(0)\\ R(0)\end{pmatrix} = \begin{pmatrix}9{,}999\\ 1\\ 0\end{pmatrix}$$

# ODEs ignore the discrete nature of individuals in a network and neglect the effects of noise.

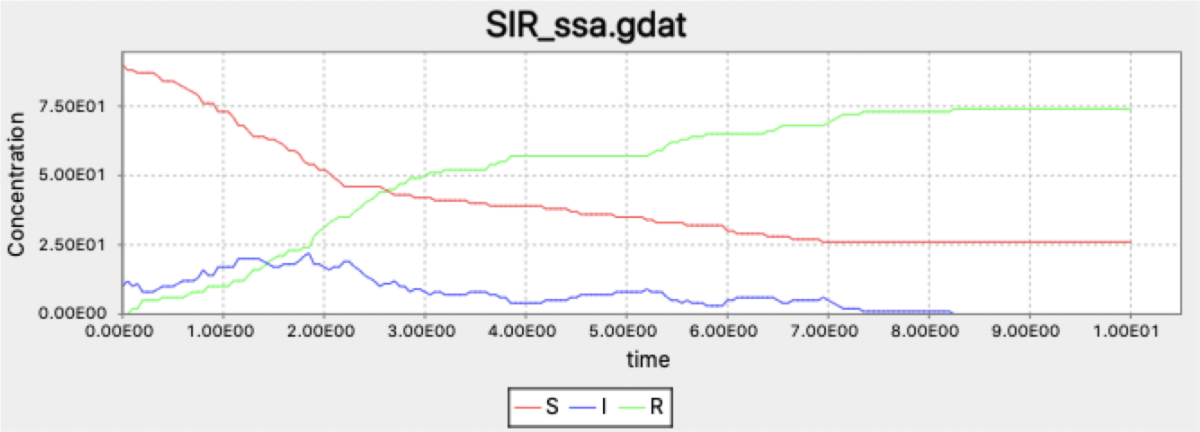
- ODEs treat all variables as continuous and cannot account for the discrete nature of individuals.
- ODEs in general (with exceptions) describe the *average* behavior of a system, but noise can affect real-world dynamics, e.g., disease outbreaks.
- The same reaction network description can be used to generate stochastic time courses (“trajectories”) that exactly sample the true probability distribution.

# Stochastic simulations can reveal the effects of noise in finite populations.

ODE simulation




Stochastic simulation




SIR model with

$$\begin{pmatrix}\beta\\ \gamma\end{pmatrix} = \begin{pmatrix}1.8\times 10^{-2}\\ 1\end{pmatrix}, \qquad \begin{pmatrix}S(0)\\ I(0)\\ R(0)\end{pmatrix} = \begin{pmatrix}90\\ 10\\ 0\end{pmatrix}$$

# Stochastic dynamics can be computed with the Gillespie Algorithm.

BioNetGen's ssa method

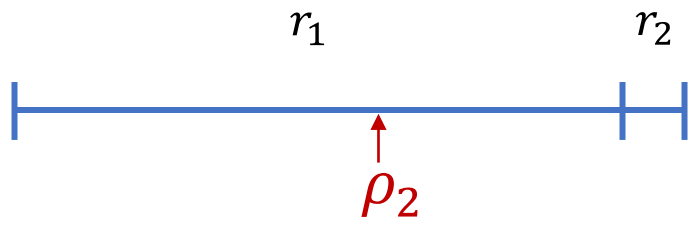
Gillespie's **Direct Method** illustrated for the SIR model. At time *t*=0 we have

$$\boldsymbol{X}(0) = \begin{pmatrix} 90 \\ 10 \\ 0 \end{pmatrix}, \quad \boldsymbol{v}(0) = \begin{pmatrix} \beta S \cdot I \\ \gamma I \end{pmatrix} = \begin{pmatrix} 16.2 \\ 1 \end{pmatrix}$$

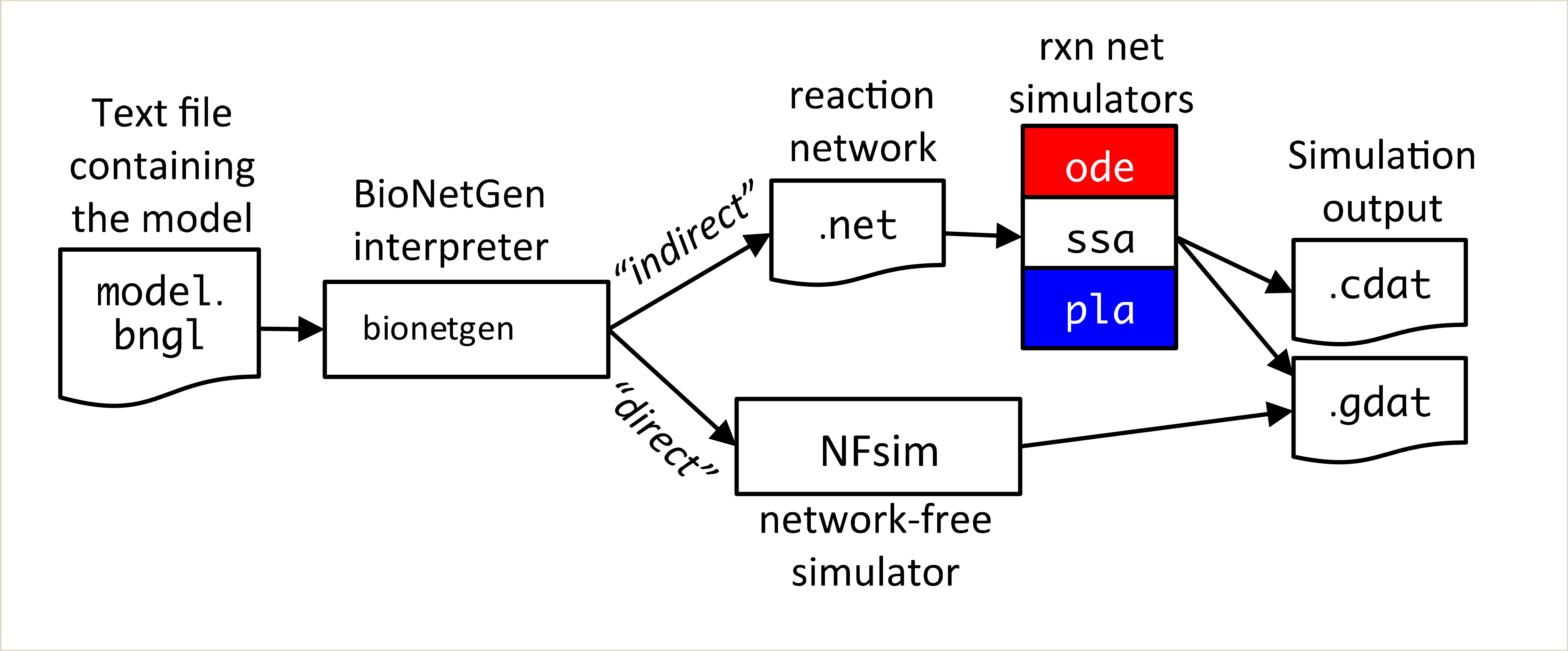
Total rate of all reactions, $a_0$, is 17.2.

1. Time of next reaction is chosen as $\tau = \dfrac{-\ln \rho_1}{a_0}$, $\quad t = t + \tau$
2. Index of next reaction, *j*, is chosen using so probability of each reaction firing is proportional to its rate.

$$\boldsymbol{X}(\tau) = \boldsymbol{X}(0) + \boldsymbol{S}_1 \qquad \boldsymbol{S}_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$$



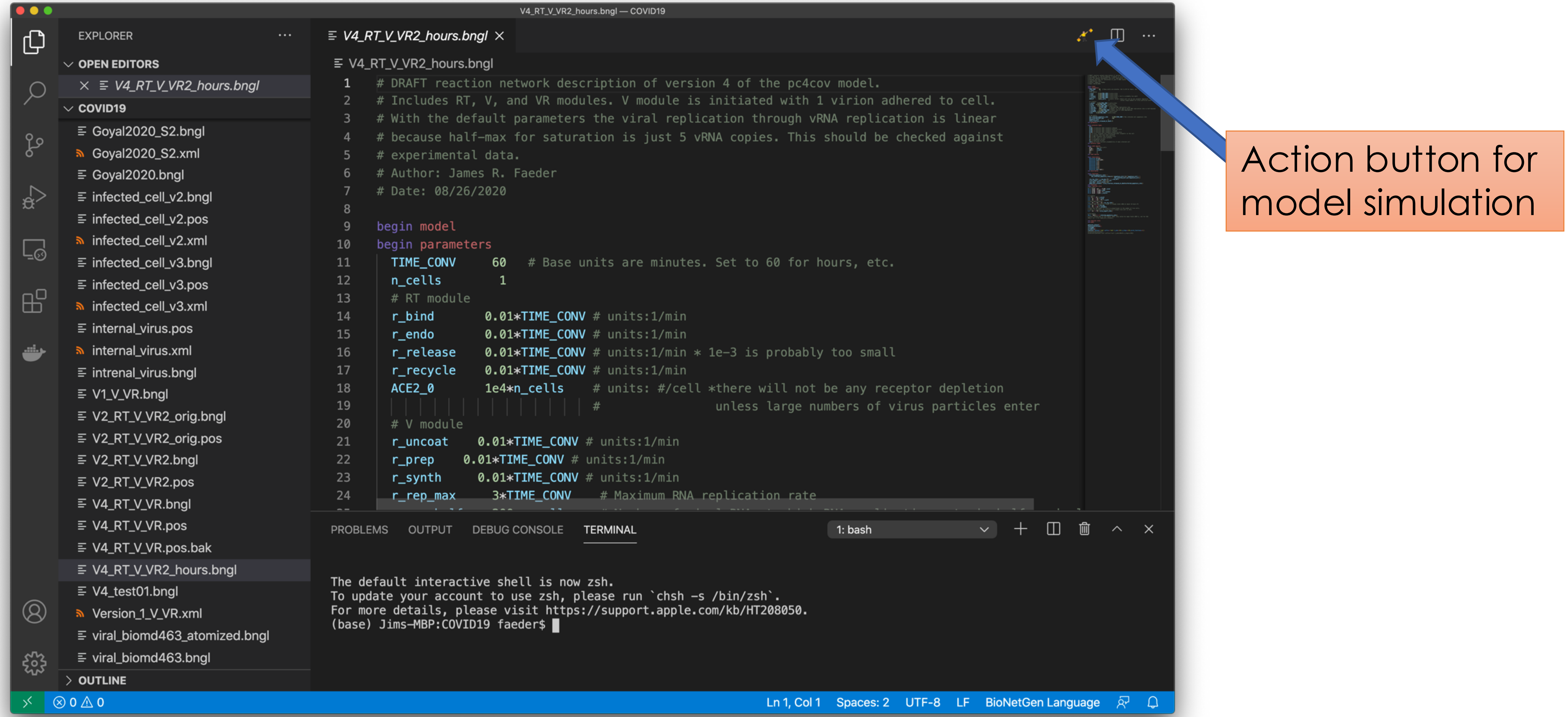
# BioNetGen enables the construction, simulation and analysis of RNMs.




Command line interface

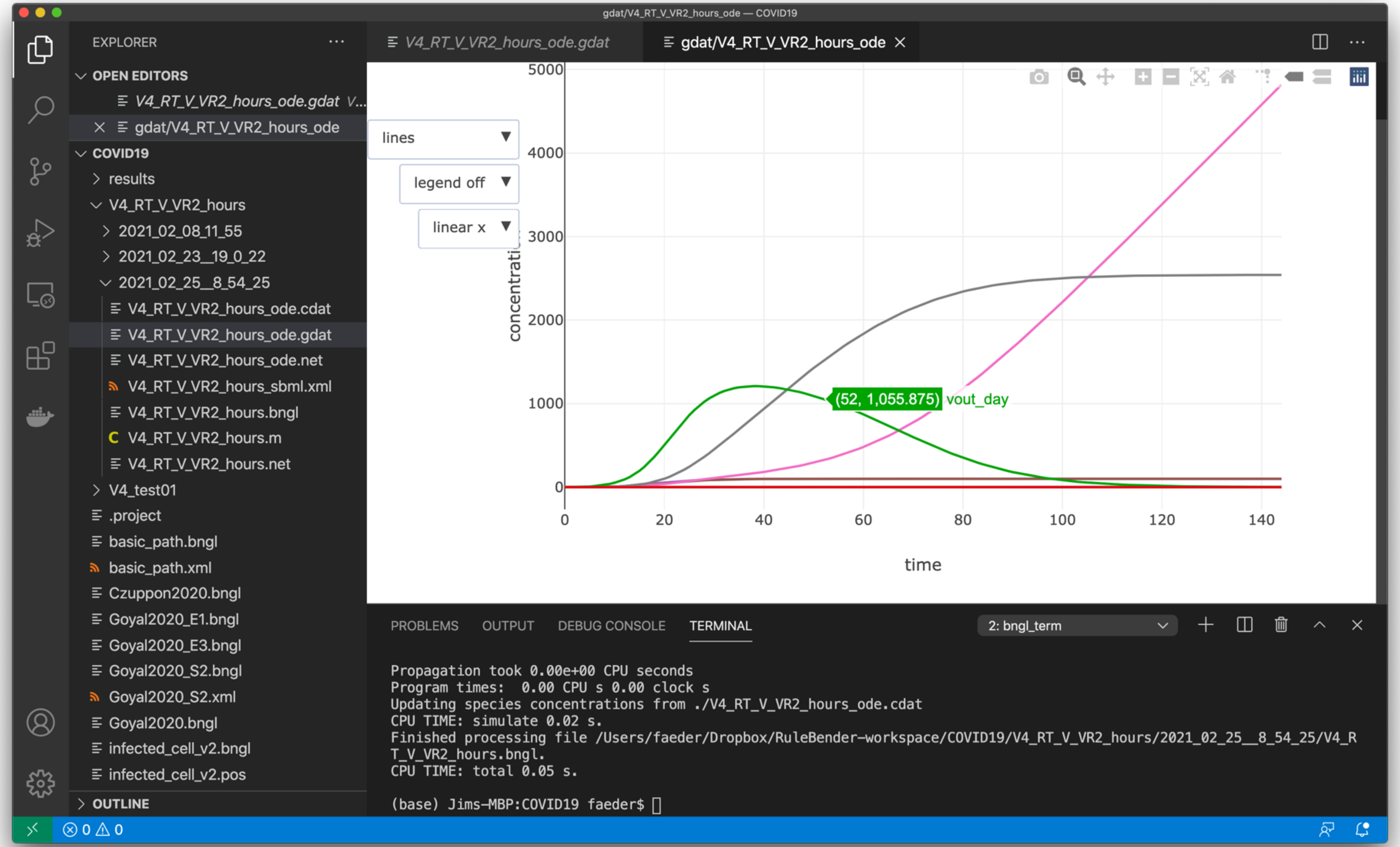
```
>bionetgen run –i model.bngl
```

# Visual Studio Code (VS Code) extension provides interactive environment for BioNetGen




Ali Sinan Saglam

# Visual Studio Code (VS Code) extension provides interactive environment for BioNetGen




Interactive plotting

Ali Sinan Saglam

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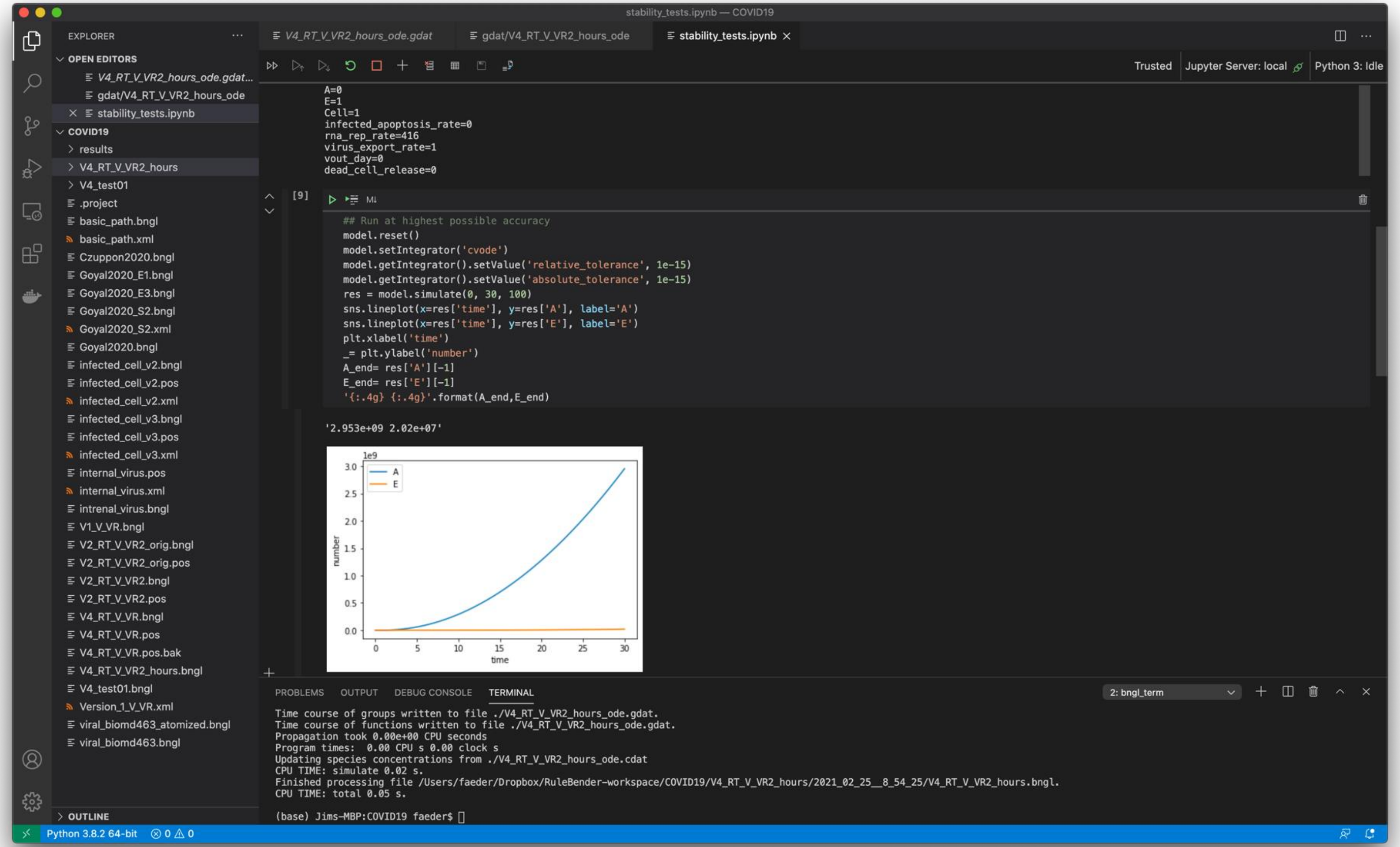



Deeper model analysis using Jupyter notebooks

# Stop here

Build SIR model using BioNetGen Extension in VS Code

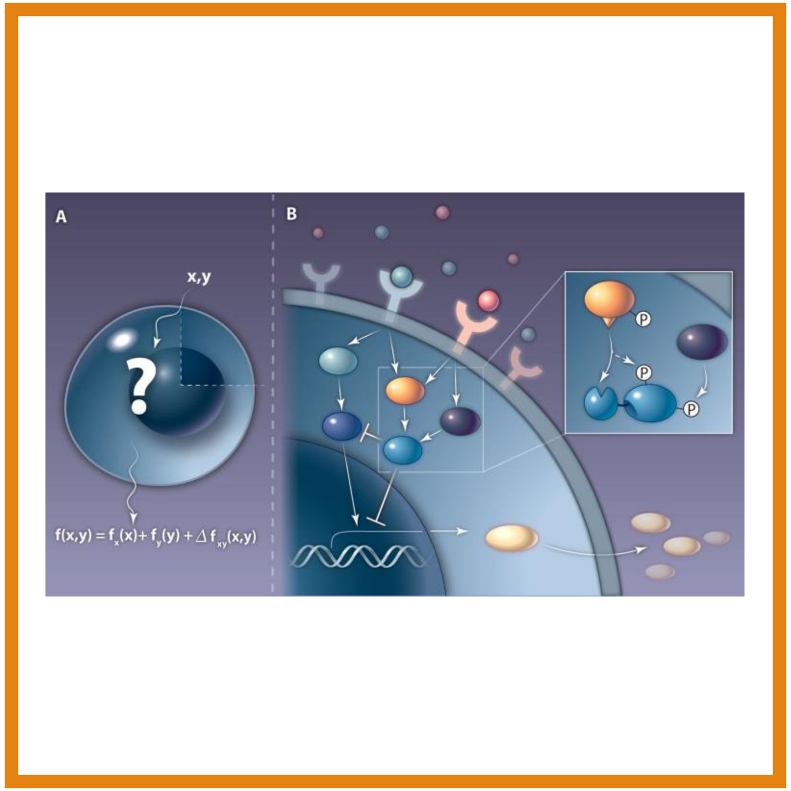
# Overview of Rule-based modeling of cell signaling

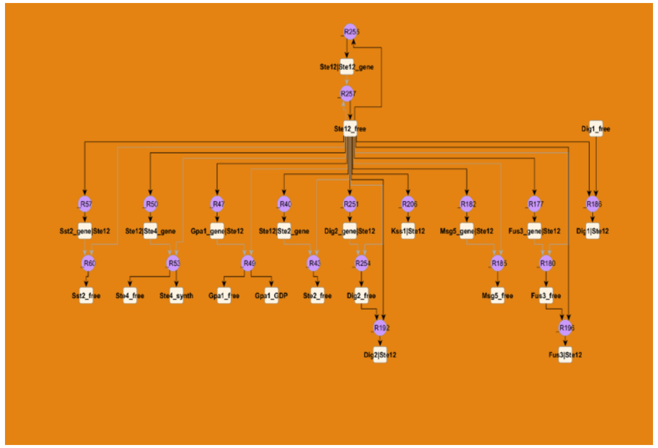

JIM FAEDER

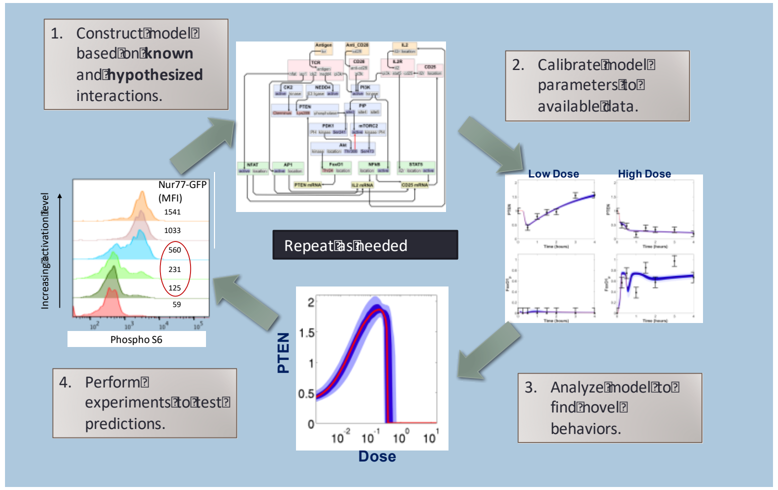
*DEPARTMENT OF COMPUTATIONAL AND SYSTEMS BIOLOGY*

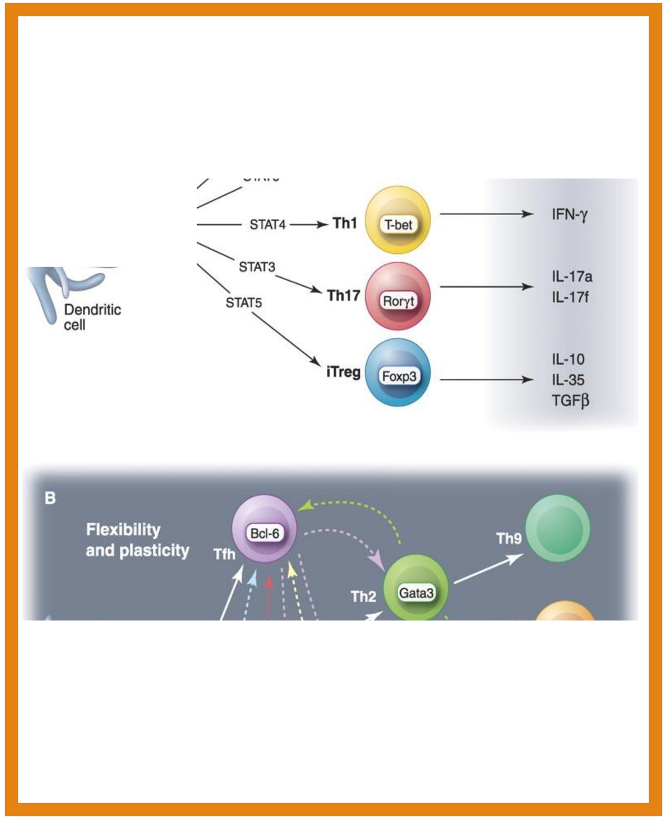
*UNIVERSITY OF PITTSBURGH SCHOOL OF MEDICINE*


MMBioS Cell Modeling Workshop
July 12, 2021

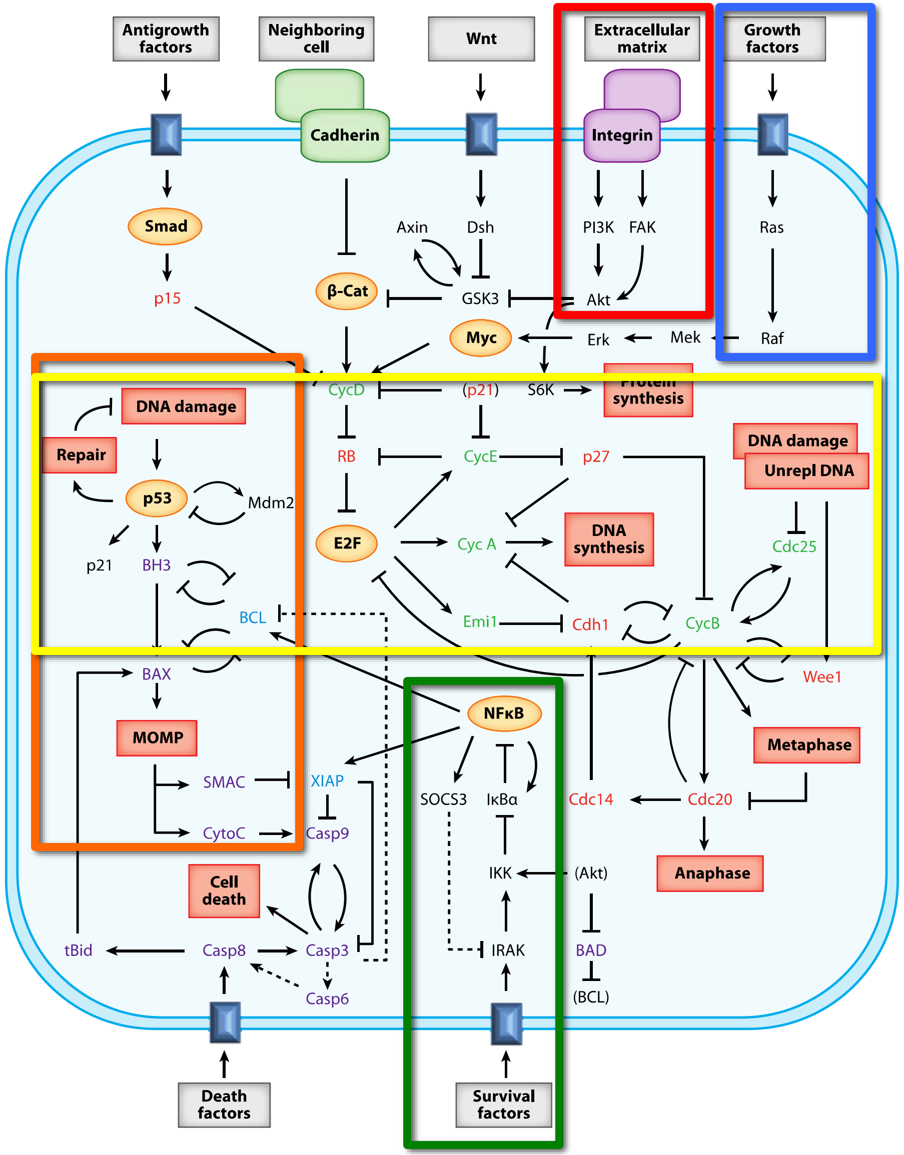




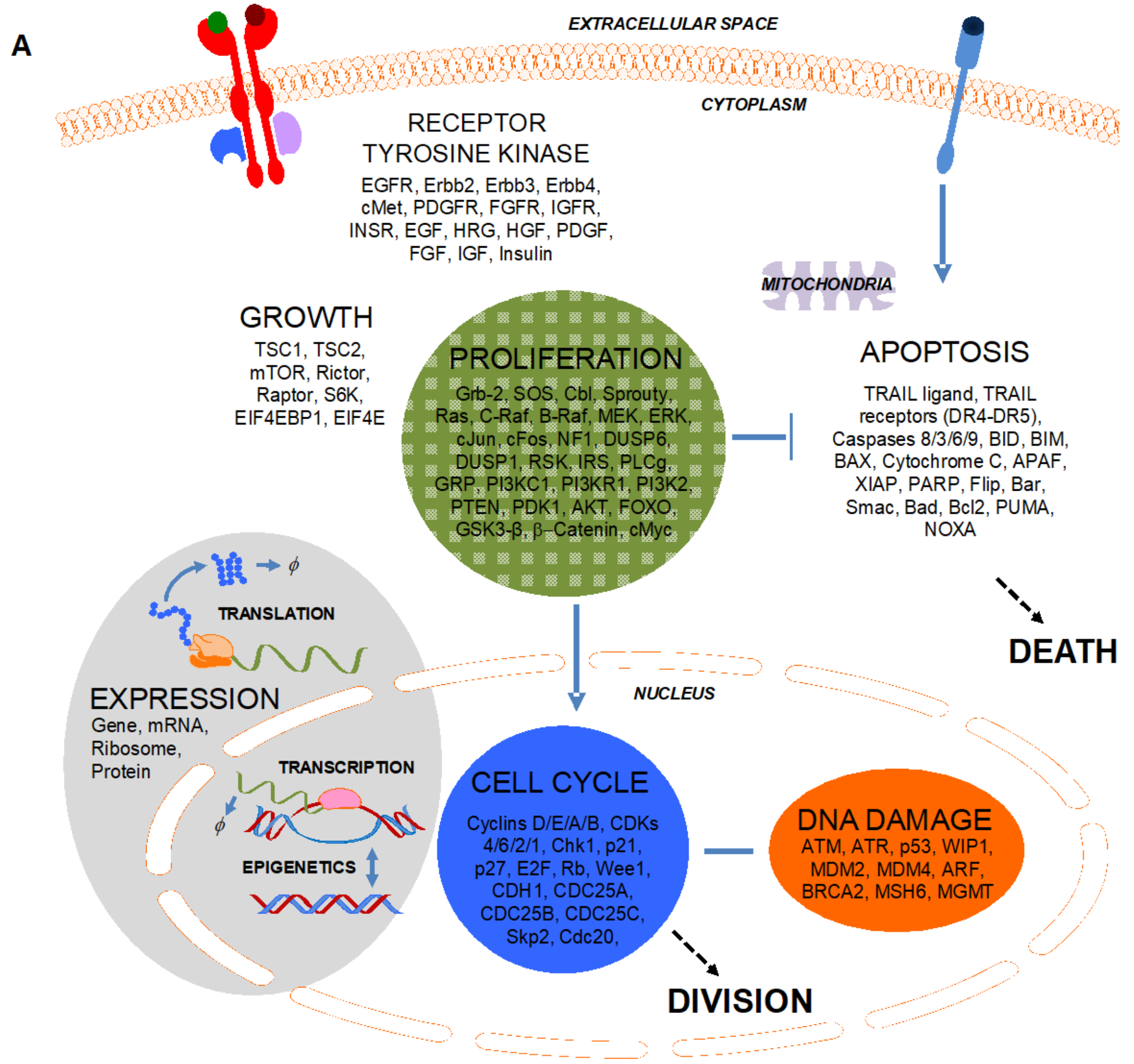



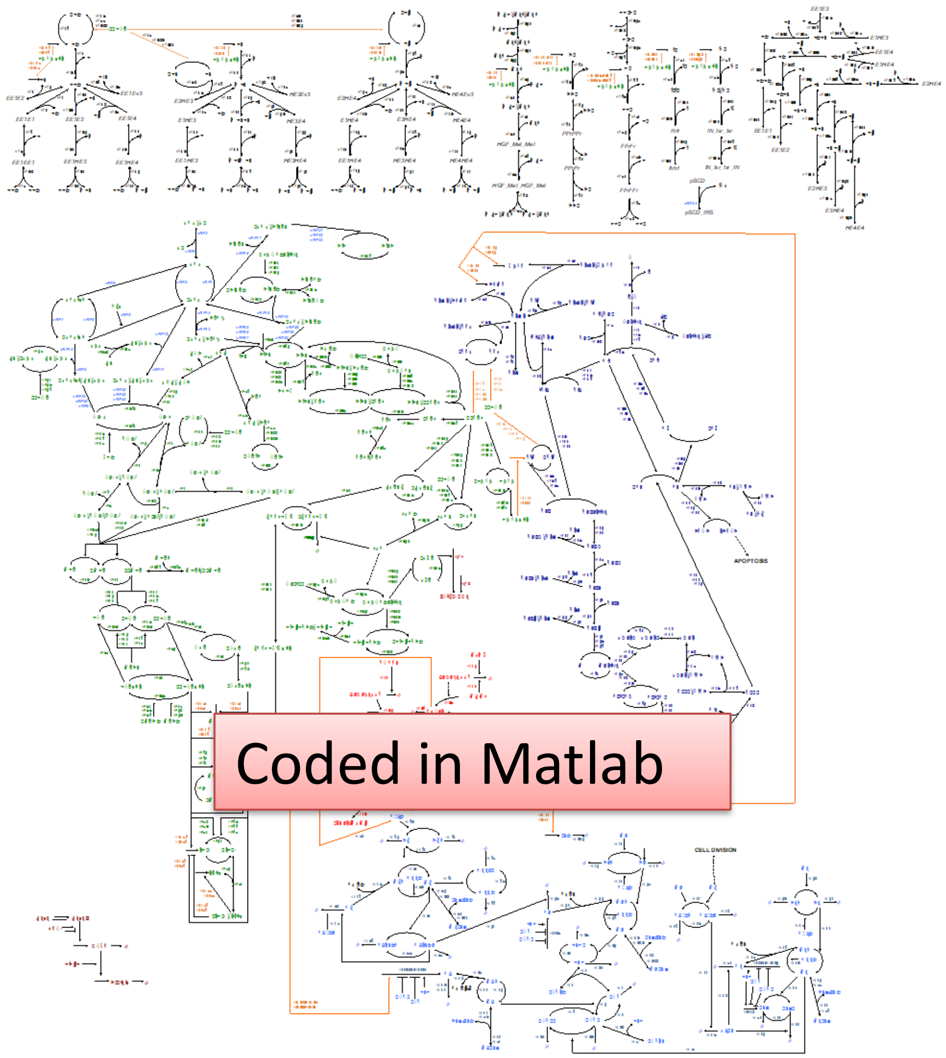
# Toward an integrated model of the cell




How to accomplish the task of linking together signaling modules that determine cell fate?

# Integrated Model of MCF10A Cell

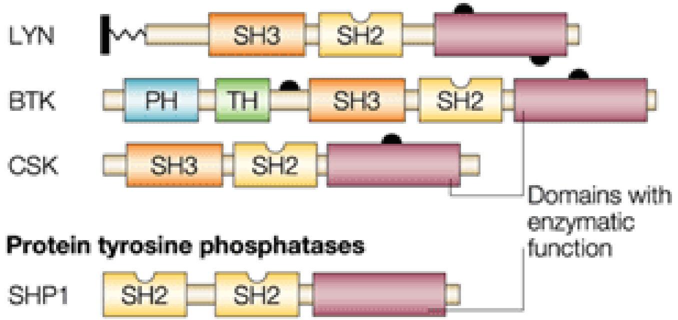



Bouhaddou et al. (2017) *PLOS Computational Biology*

# Challenges of modeling cell regulatory networks

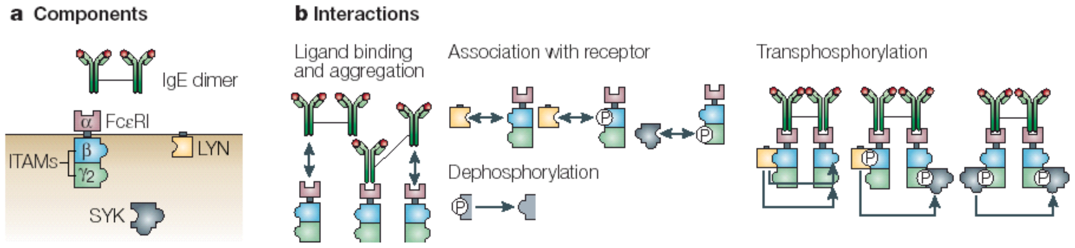
- Proteins are multi-functional

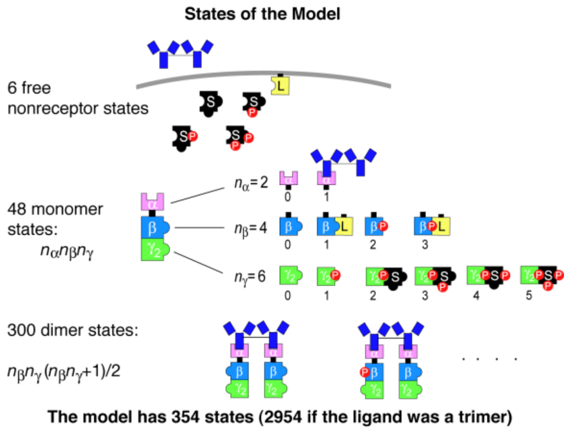



multiple sites of binding

multiple sites of posttranslational modification

- Representing their known interactions requires handling of *combinatorial complexity*




**Small number of rules**

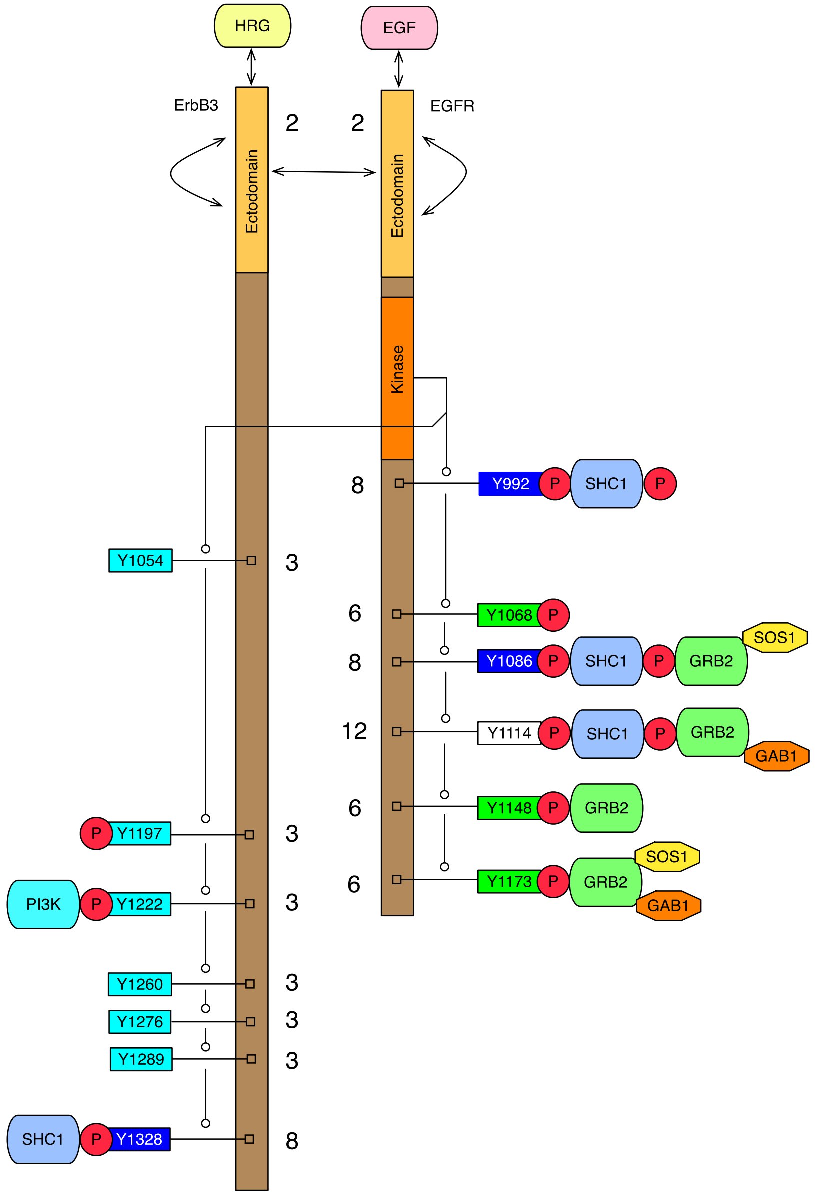
Small number of components and interactions ➔huge number of possible species and reactions

# Combinatorial complexity in a realistic model of EGFR signaling

ErbB3:ErbB1 has > $3.8\times10^{9}$ states

ErbB1:ErbB1 has > $5.5\times10^{10}$ states

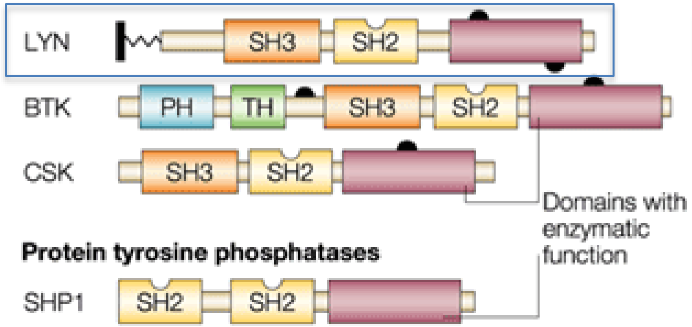
**Such models can be built from a manageable number of rules – 10's to 100's**




Creamer et al. (2012) *BMC Syst. Biol.*

# What is Rule-based Modeling (RBM)?

Molecules are modeled as *structured objects*

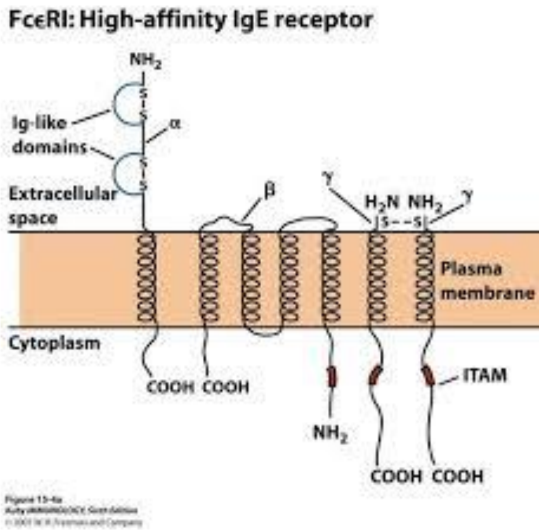



LYN(SH3,SH2,kin,Y397~0~P,Y508~0~P)

Molecule

Components

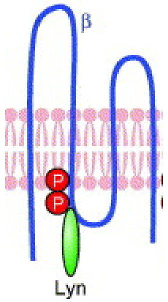
States




FceRI(a_Ig,b_Y218~0~P,g_ITAM~0~P)

# What is Rule-based Modeling (RBM)?

## Rules define the interactions of molecules




"Lyn SH2 domain binds to phosphorylated Tyr 218 on the β subunit of FcεRI "

**Rule:** `LYN(SH2) + FceRI(bY_218~P) <-> LYN(SH2!1).FceRI(bY_218~P!1) kpL, kmL`

**Reactants:** `LYN(SH2) + FceRI(bY_218~P)`

**Products:** `LYN(SH2!1).FceRI(bY_218~P!1)`

**Rate Law:** `kpL, kmL`

**bond:** `!1`

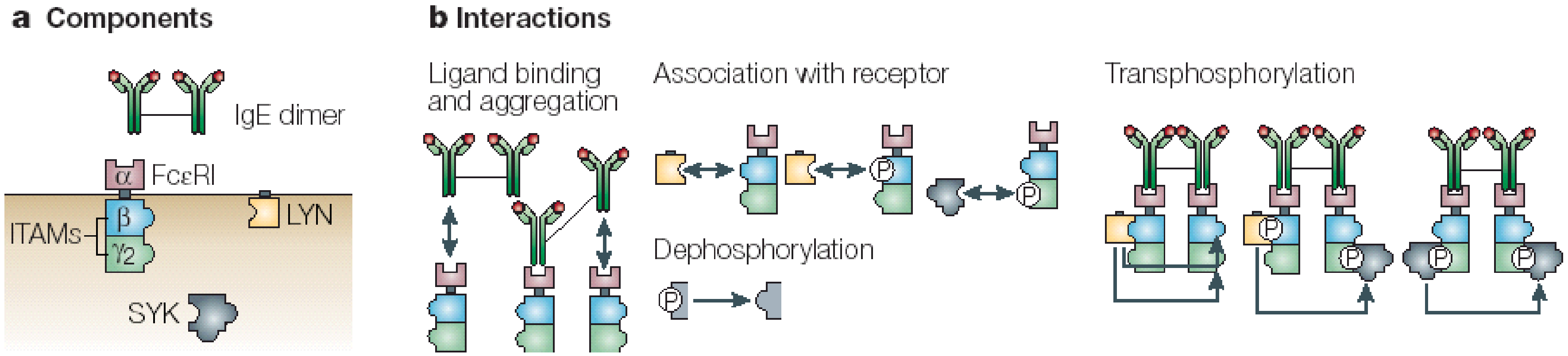
"Don't write don't care" – elements not mentioned may be in any state

➔ *One rule can generate reactions involving many different species*

Reaction rate determined by **Mass Action kinetics**

rate forward = `kpL`*[`Lyn(SH2)`]*[`FceRI(bY_218~P)`]

# Composition of a Rule-Based Model




## Molecules (4)

```
begin molecule types
Lig(l,l)
Lyn(U,SH2)
Syk(tSH2,l~U~P,a~U~P)
Rec(a,b~U~P,g~U~P)
end molecule types
```

## Reaction Rules (19)

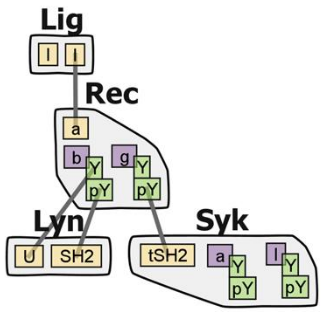
BioNetGen language

```
begin reaction_rules
# Ligand-receptor binding
1 Rec(a) + Lig(l,l) <-> Rec(a!1).Lig(l!1,l) kp1, km1
 Rec(a) + Lig(l,l) <-> Rec(a!1).Lig(l!1,l) kp1, km1

# Receptor-aggregation
2 Rec(a) + Lig(l,l!1) <-> Rec(a!2).Lig(l!2,l!1) kp2,km2

# Constitutive Lyn-receptor binding
3 Rec(b~Y) + Lyn(U,SH2) <-> Rec(b~Y!1).Lyn(U!1,SH2) kpL, kmL
…
```

# Composition of a Rule-Based Model




**Molecule types (4)**

**Reaction Rules (19)**

Network Generation

**Species (354)**

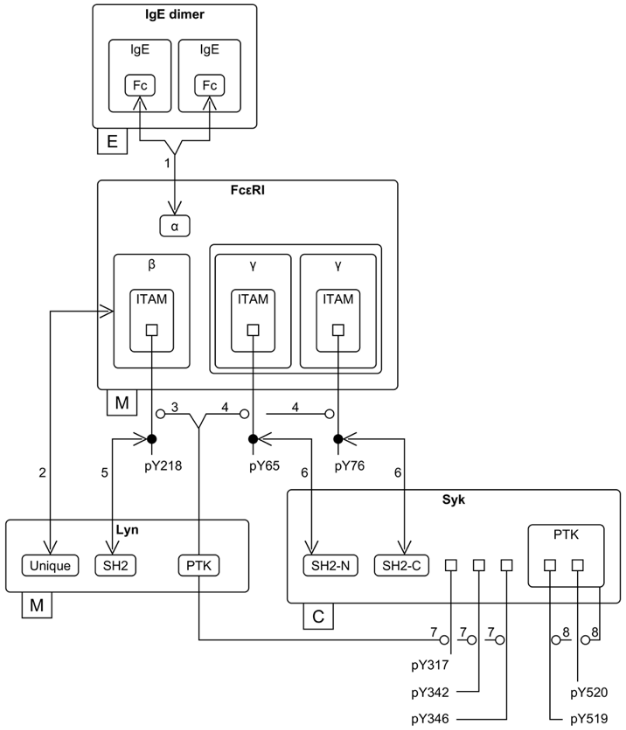
**Reactions(3680)**

1 Lig(l,l) Lig_tot
2 Lyn(SH2,U) Lyn_tot
3 Syk(a~Y,l~Y,tSH2) Syk_tot
4 Rec(a,b~Y,g~Y) Rec_tot
5 Lig(l!1,l).Rec(a!1,b~Y,g~Y) 0
6 Lyn(SH2,U!1).Rec(a,b~Y!1,g~Y) 0
7 Lig(l!1,l).Lyn(SH2,U!2).Rec(a!1,b~Y!2,g~Y) 0
8 Lig(l!1,l!2).Rec(a!2,b~Y,g~Y).Rec(a!1,b~Y,g~Y) 0
9 Lig(l!1,l!2).Lyn(SH2,U!3).Rec(a!2,b~Y!3,g~Y).Rec(a!1,b~Y,g~Y) 0
...

1 Rec(a,b~Y,g~Y) + Lig(l,l) -> Lig(l!1,l).Rec(a!1,b~Y,g~Y) 2*kp1
2 Rec(a,b~Y,g~Y) + Lyn(SH2,U) -> Lyn(SH2,U!1).Rec(a,b~Y!1,g~Y) kpL
3 Lyn(SH2,U!1).Rec(a,b~Y!1,g~Y) + Lig(l,l) -> Lig(l!1,l).Lyn(SH2,U!2).Rec(a!1,b~Y!2,g~Y) 2*kp1
4 Lig(l!1,l).Rec(a!1,b~Y,g~Y) -> Rec(a,b~Y,g~Y) + Lig(l,l) km1
5 Rec(a,b~Y,g~Y) + Lig(l!1,l).Rec(a!1,b~Y,g~Y) -> Lig(l!1,l!2).Rec(a!2,b~Y,g~Y).Rec(a!1,b~Y,g~Y) kp2
6 Lyn(SH2,U!1).Rec(a,b~Y!1,g~Y) + Lig(l!1,l).Rec(a!1,b~Y,g~Y) -> Lig(l!1,l!2).Lyn(SH2,U!3).Rec(a!2,b~Y!3,g~Y).Rec(a!1,b~Y,g~Y) kp2 7 Lig(l!1,l).Rec(a!1,b~
kpL
8 Lyn(SH2,U!1).Rec(a,b~Y!1,g~Y) -> Rec(a,b~Y,g~Y) + Lyn(SH2,U) kmL
9 Lig(l!1,l).Lyn(SH2,U!2).Rec(a!1,b~Y!2,g~Y) -> Lyn(SH2,U!1).Rec(a,b~Y!1,g~Y) + Lig(l,l) km1
...

# Rule-based modeling enables knowledge representation on a large scale

## FcεRI model

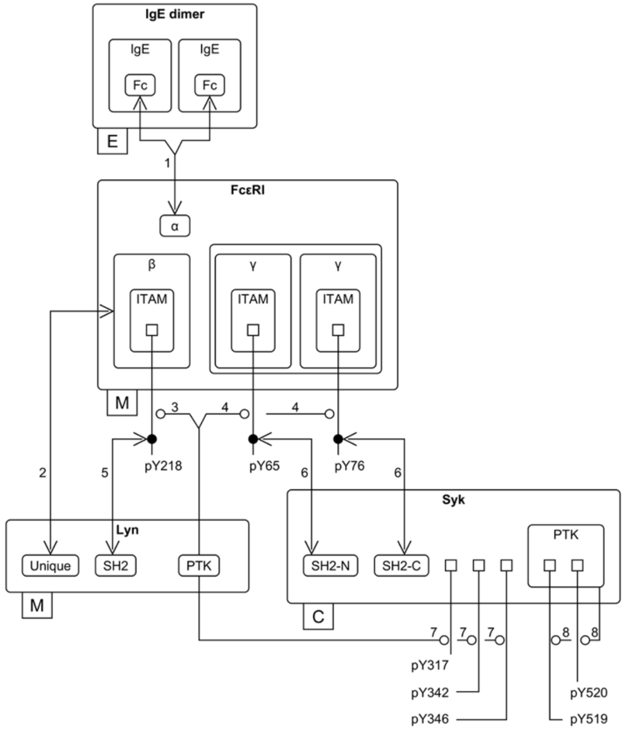



- Precise encoding of modeled structures and interactions
- User avoids combinatorial complexity
- Amenable to visualization
- Extensible as knowledge base grows

Faeder et al., *J. Immunol.* (2003)
Chylek et al., *Mol. BioSys.* (2011)

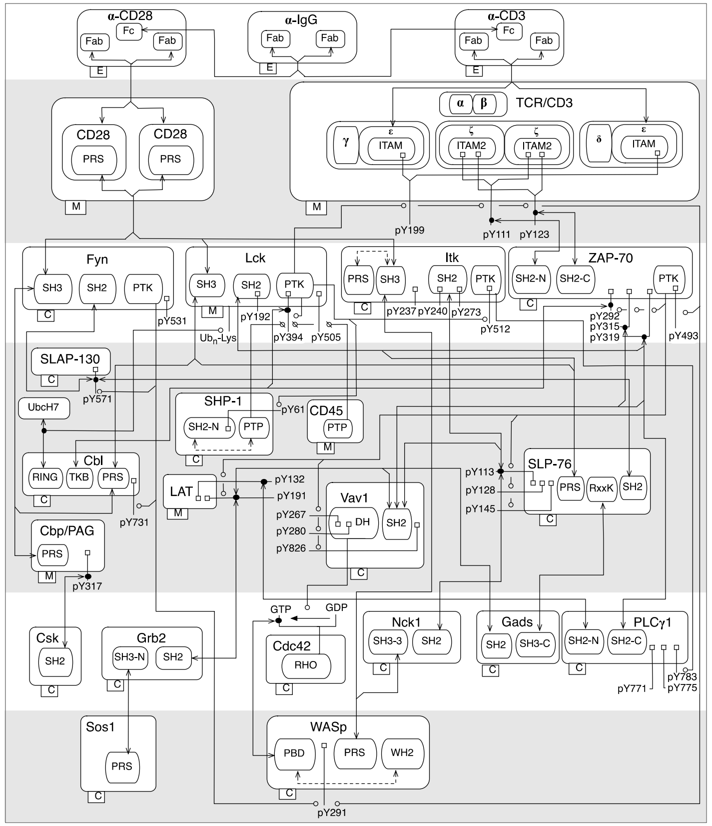
# Rule-based modeling enables knowledge representation on a large scale

## FcεRI model




Faeder et al., *J. Immunol.* (2003)

## T cell receptor model

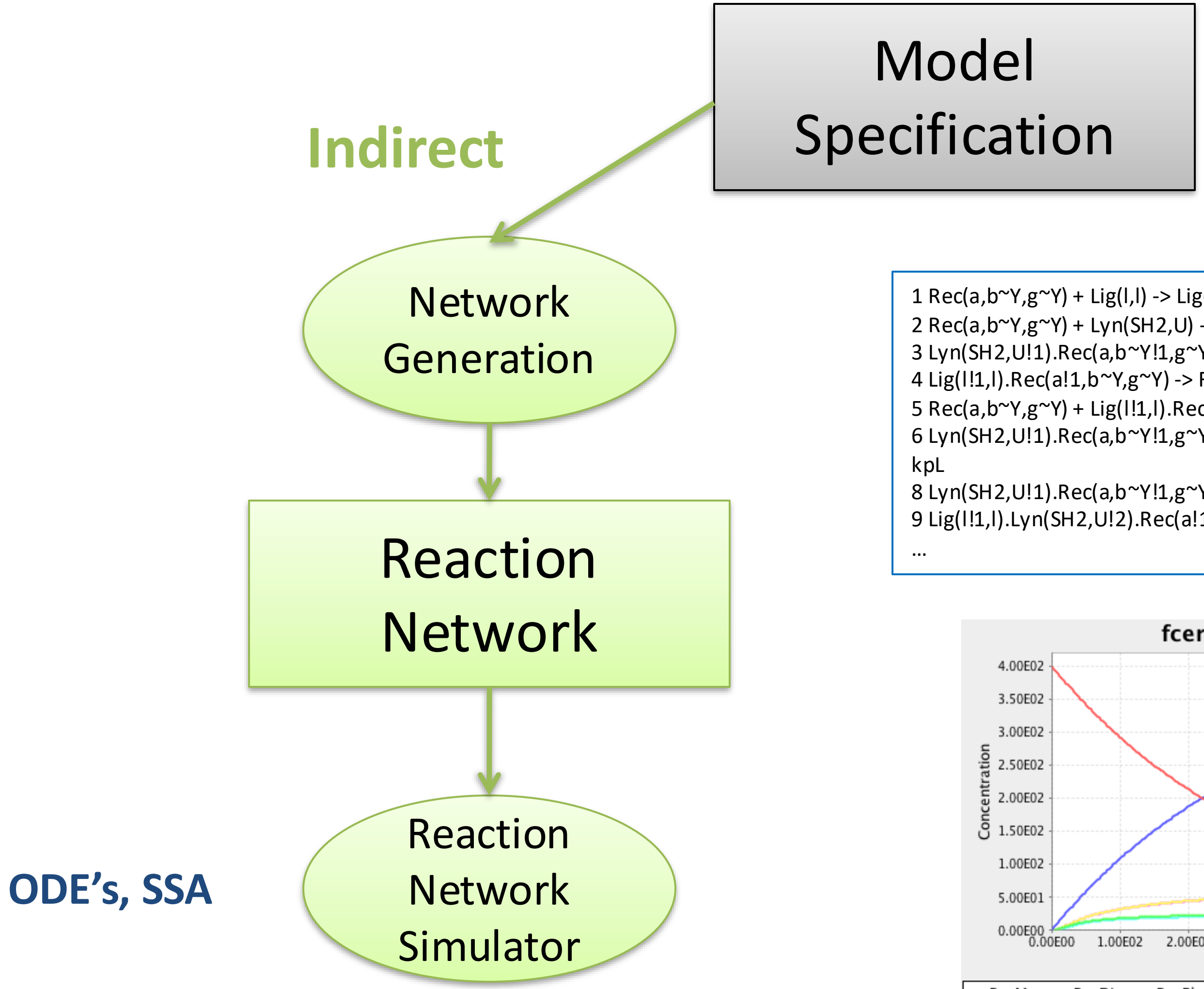



Chylek et al., *PLOS One* (2014)

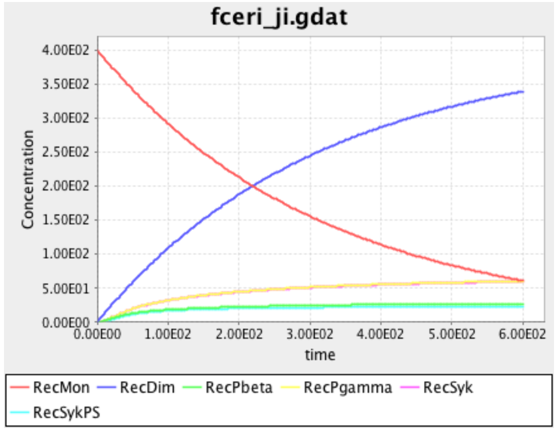
**20 molecule types**
**129 rules**

***Number of species is too large for network generation!***

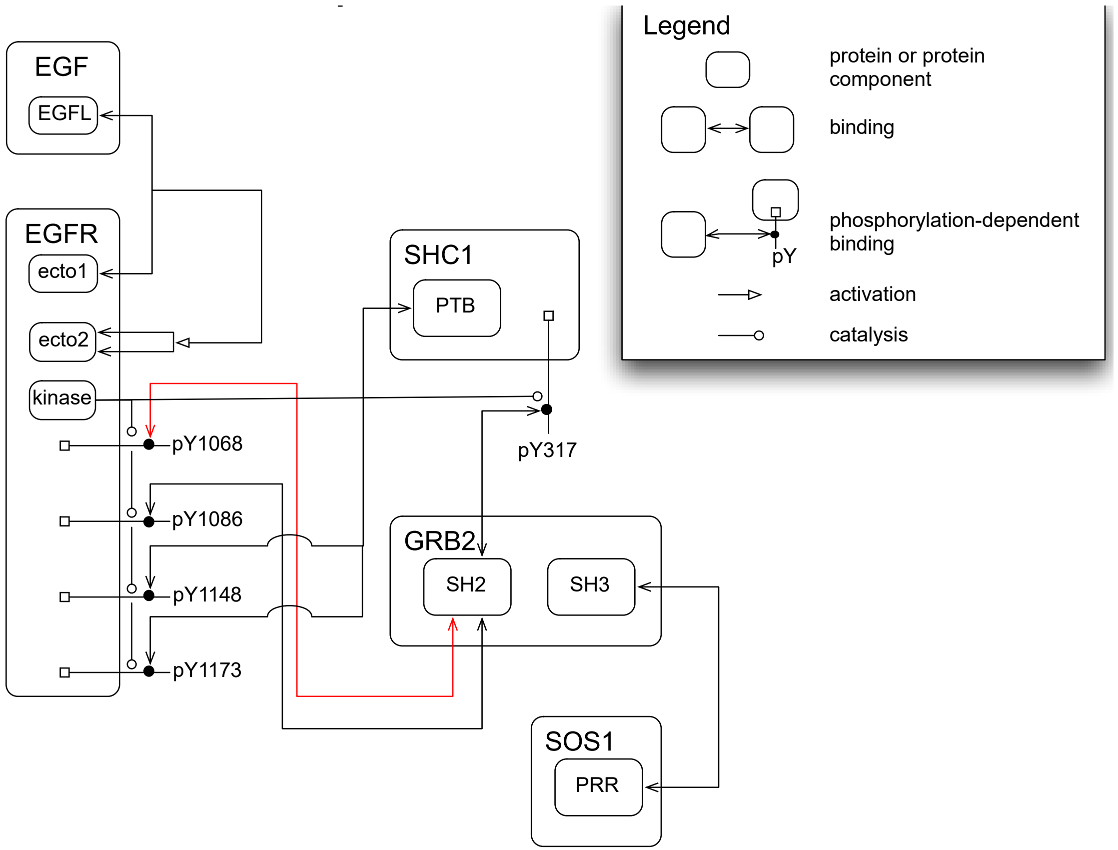
# Methods for simulating RBMs




```
1 Rec(a,b~Y,g~Y) + Lig(l,l) -> Lig(l!1,l).Rec(a!1,b~Y,g~Y) 2*kp1
2 Rec(a,b~Y,g~Y) + Lyn(SH2,U) -> Lyn(SH2,U!1).Rec(a,b~Y!1,g~Y) kpL
3 Lyn(SH2,U!1).Rec(a,b~Y!1,g~Y) + Lig(l,l) -> Lig(l!1,l).Lyn(SH2,U!2).Rec(a!1,b~Y!2,g~Y) 2*kp1
4 Lig(l!1,l).Rec(a!1,b~Y,g~Y) -> Rec(a,b~Y,g~Y) + Lig(l,l) km1
5 Rec(a,b~Y,g~Y) + Lig(l!1,l).Rec(a!1,b~Y,g~Y) -> Lig(l!1,l!2).Rec(a!2,b~Y,g~Y).Rec(a!1,b~Y,g~Y) kp2
6 Lyn(SH2,U!1).Rec(a,b~Y!1,g~Y) + Lig(l!1,l).Rec(a!1,b~Y,g~Y) -> Lig(l!1,l!2).Lyn(SH2,U!3).Rec(a!2,b~Y!3,g~Y).Rec(a!1,b~Y,g~Y) kp2 7 Lig(l!1
kpL
8 Lyn(SH2,U!1).Rec(a,b~Y!1,g~Y) -> Rec(a,b~Y,g~Y) + Lyn(SH2,U) kmL
9 Lig(l!1,l).Lyn(SH2,U!2).Rec(a!1,b~Y!2,g~Y) -> Lyn(SH2,U!1).Rec(a,b~Y!1,g~Y) + Lig(l,l) km1
...
```



# Combinatorial Complexity




5 proteins, 20 interactions → 170,000 species

# Methods for simulating RBMs

Model Specification

**Indirect**

Network Generation

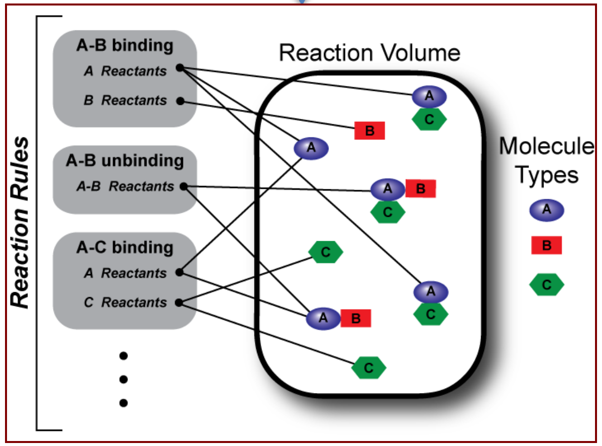
Reaction Network

Reaction Network Simulator

**Direct**

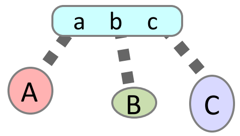
Configuration Generation

Bypasses network generation



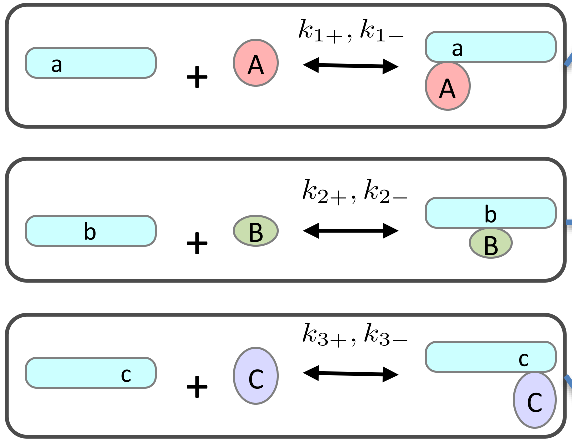

**NFsim**

# Indirect Methods – Network Generation




contact map

**reaction rules**




$$
\begin{aligned}
\mathtt{S} + \mathtt{A} &\underset{}{\overset{k_{1+},k_{1-}}{\rightleftharpoons}} \mathtt{SA}\\
\mathtt{SB} + \mathtt{A} &\overset{k_{1+},k_{1-}}{\rightleftharpoons} \mathtt{SAB}\\
\mathtt{SC} + \mathtt{A} &\overset{k_{1+},k_{1-}}{\rightleftharpoons} \mathtt{SAC}\\
\mathtt{SBC} + \mathtt{A} &\overset{k_{1+},k_{1-}}{\rightleftharpoons} \mathtt{SABC}
\end{aligned}
$$

$$
\begin{aligned}
\mathtt{S} + \mathtt{B} &\overset{k_{2+},k_{2-}}{\rightleftharpoons} \mathtt{SB}\\
\mathtt{SA} + \mathtt{B} &\overset{k_{2+},k_{2-}}{\rightleftharpoons} \mathtt{SAB}\\
\mathtt{SC} + \mathtt{B} &\overset{k_{2+},k_{2-}}{\rightleftharpoons} \mathtt{SBC}\\
\mathtt{SAC} + \mathtt{B} &\overset{k_{2+},k_{2-}}{\rightleftharpoons} \mathtt{SABC}
\end{aligned}
$$

$$
\begin{aligned}
\mathtt{S} + \mathtt{C} &\overset{k_{3+},k_{3-}}{\rightleftharpoons} \mathtt{SC}\\
\mathtt{SA} + \mathtt{C} &\overset{k_{3+},k_{3-}}{\rightleftharpoons} \mathtt{SAC}\\
\mathtt{SB} + \mathtt{C} &\overset{k_{3+},k_{3-}}{\rightleftharpoons} \mathtt{SBC}\\
\mathtt{SAB} + \mathtt{C} &\overset{k_{3+},k_{3-}}{\rightleftharpoons} \mathtt{SABC}
\end{aligned}
$$

**reactions**

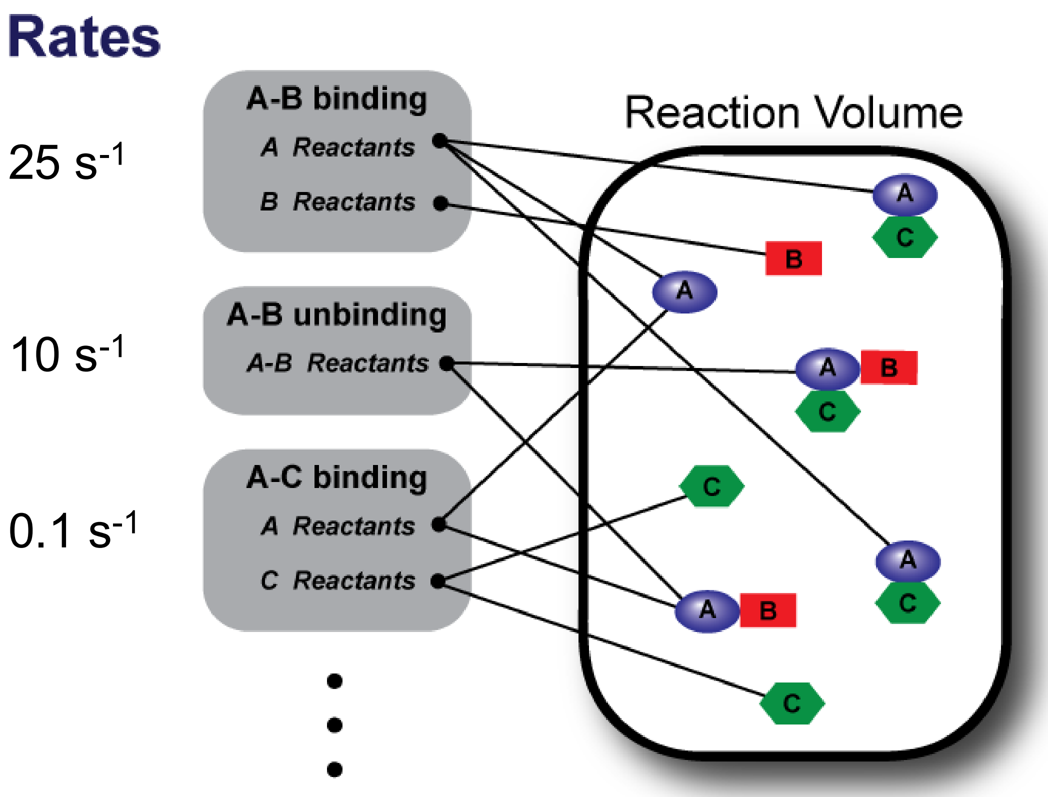
S, A, B, C,
SA, SB, SC,
SAB, SAC, SBC,
SABC

**species**

4 molecule types
3 rules

11 species
12 reactions

# NFsim Algorithm

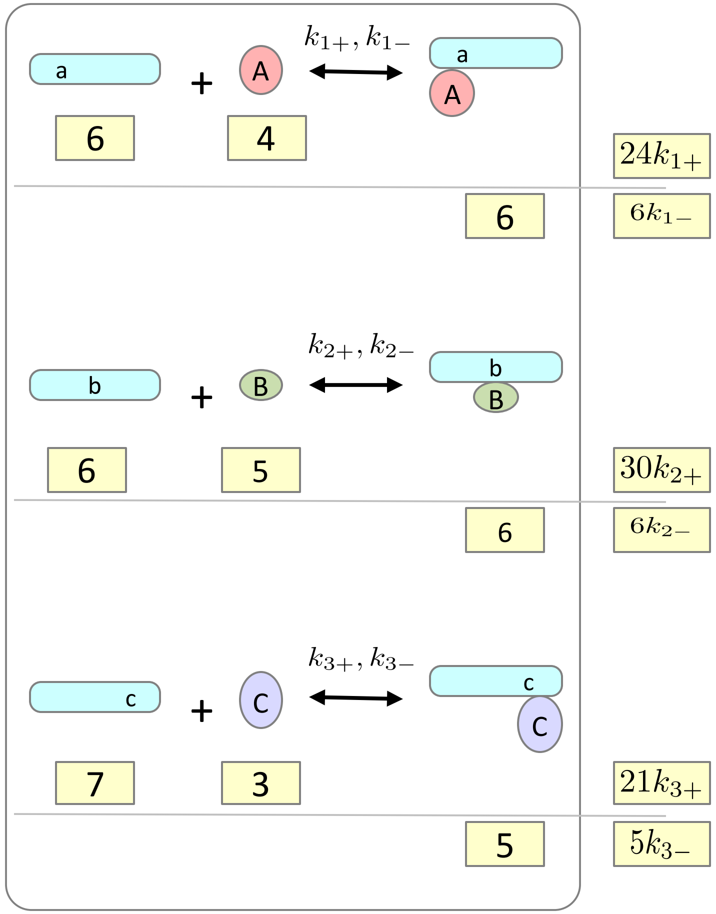



0. **Initialize *reactant lists* and calculate *rule propensities.***
1. Select next reaction time and next *rule*.
2. Select molecules and sites to react.
   a. Check any application condition(s).
3. Apply operation specified by rule.
4. Update reactant lists and propensities.
5. Increment time.

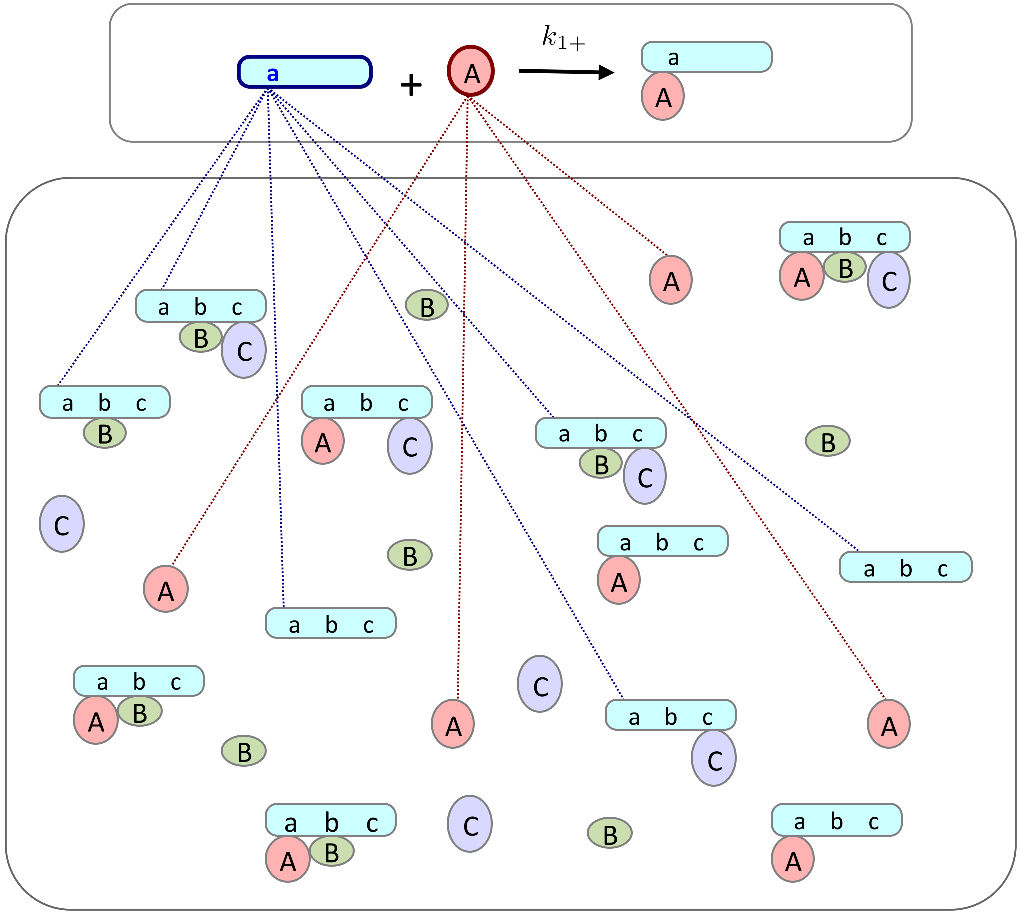
Sneddon, Faeder, and Emonet (2011) *Nature Methods*

# Direct Methods (NFsim)

## reaction rules



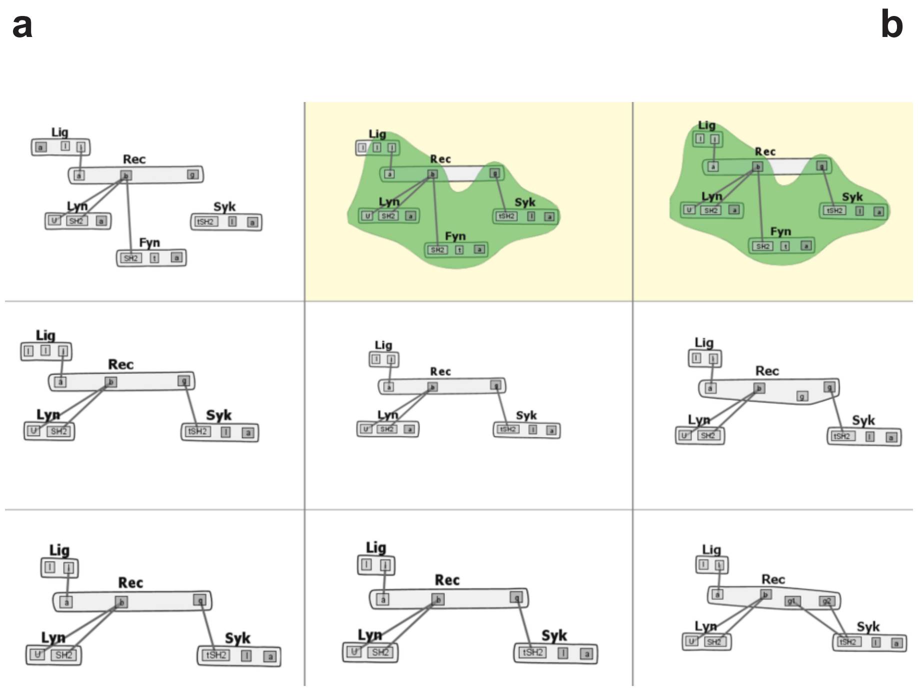

## event generation

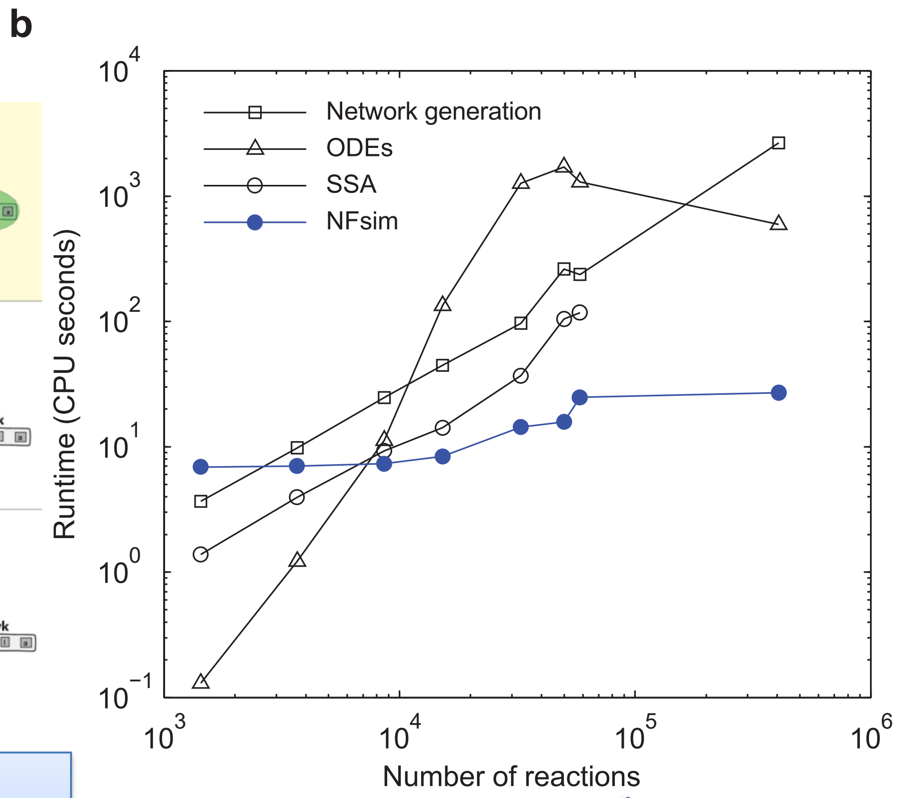



## system configuration

*total propensity* $24k_{1+} + 6k_{1-} + 30k_{2+} + 6k_{2-} + 21k_{3+} + 5k_{3-}$

# FcεRI signaling models



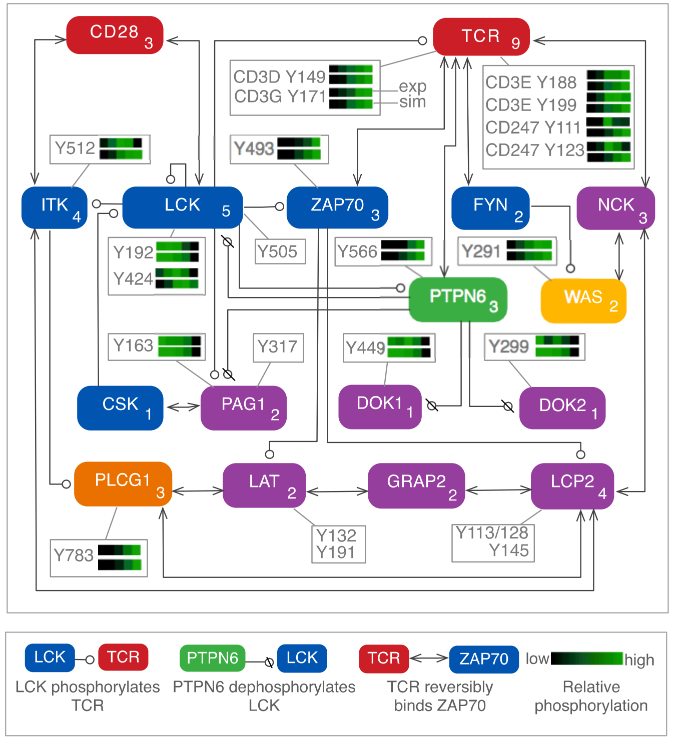

Increasing complexity

NFSIM can simulate models of greatly increased complexity with manageable increase in cost.

MW Sneddon, JRF, and T. Emonet (2011) *Nat. Methods*, **8**, 177

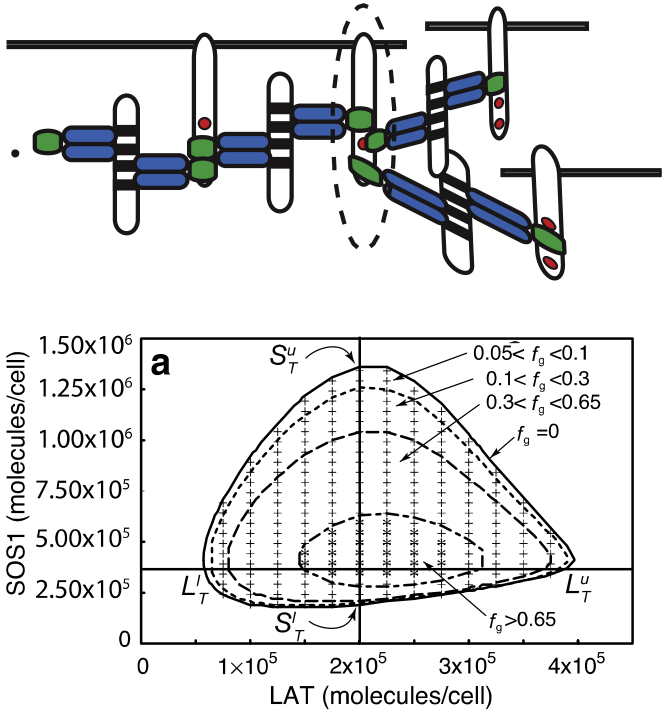
# Other examples of network-free applications

### Phosphoproteome dynamics



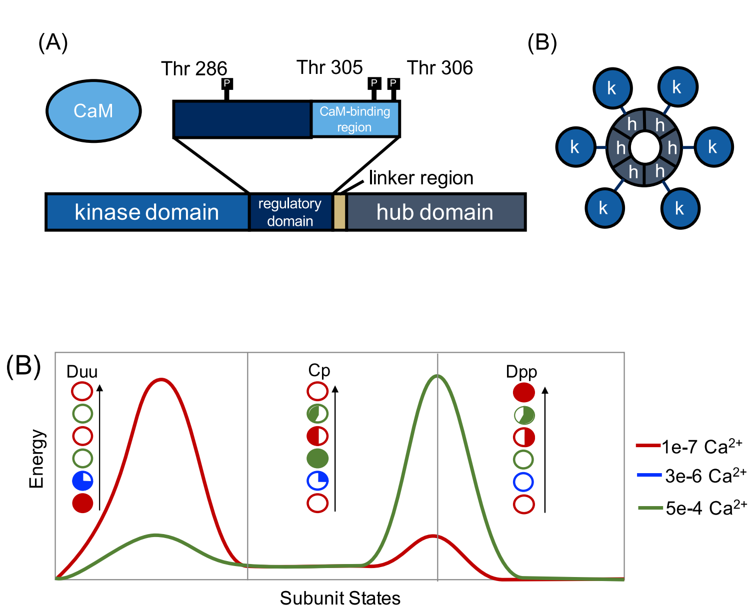
Chylek et al. (2014) *PLOS One*

### Multivalent aggregation in signaling



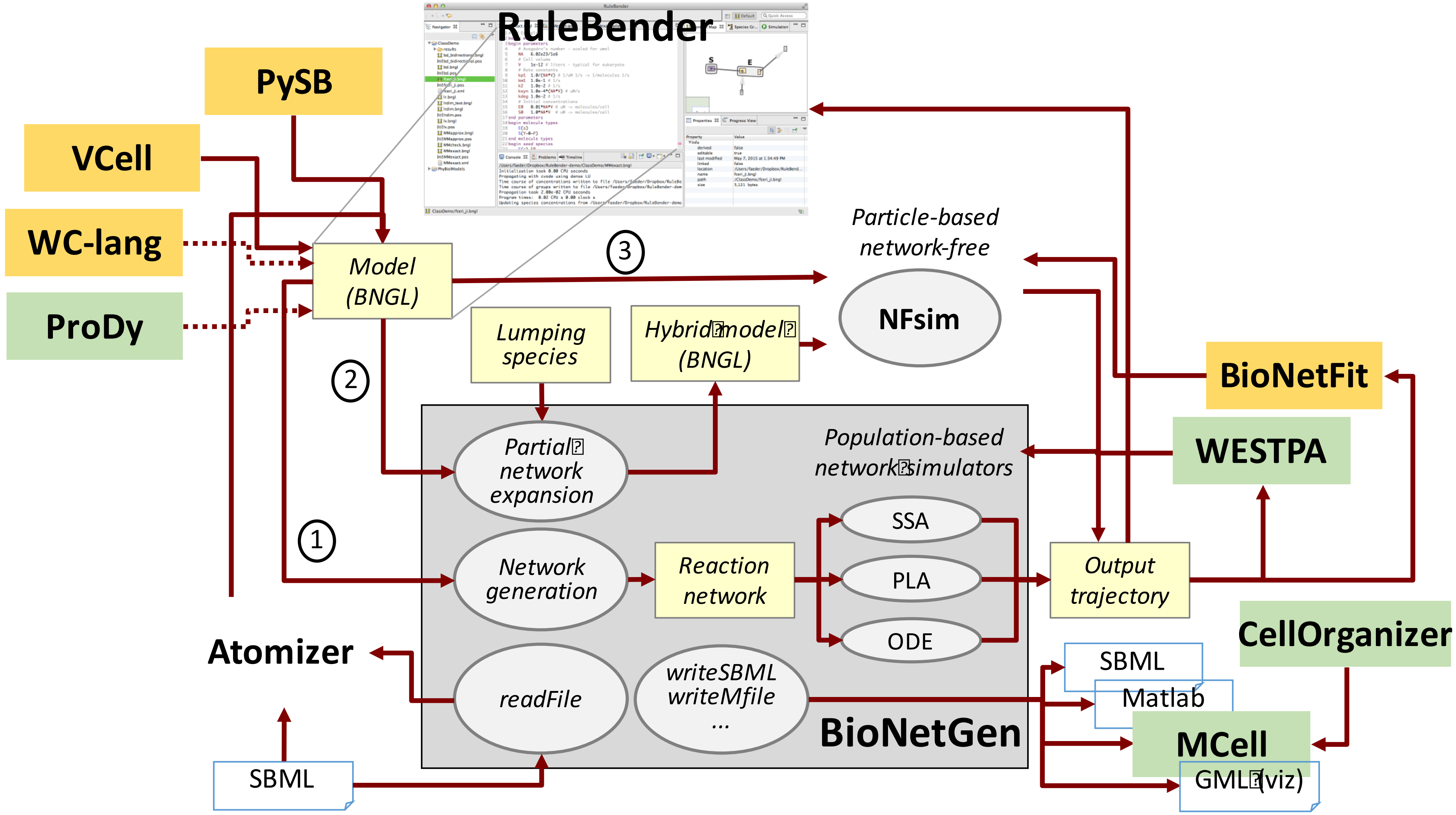
Nag et al. (2009) *Biophys. J*

### Holoenzyme dynamics



Kaya, Grundy et al., in progress.

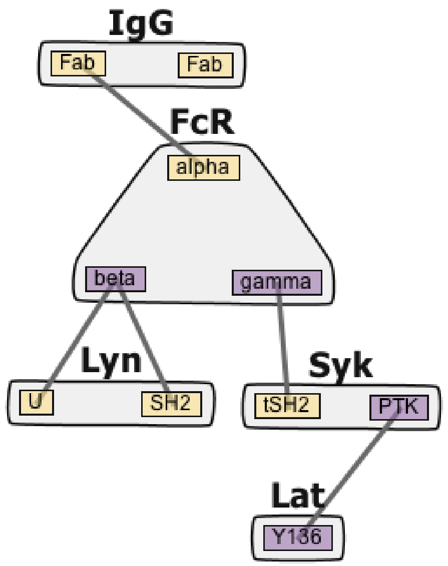
# BioNetGen and associated modeling tools




Harris, Hogg, et al. (2016) *Bioinformatics*, 32 (21): 3366-3368.

# Why use RBM?

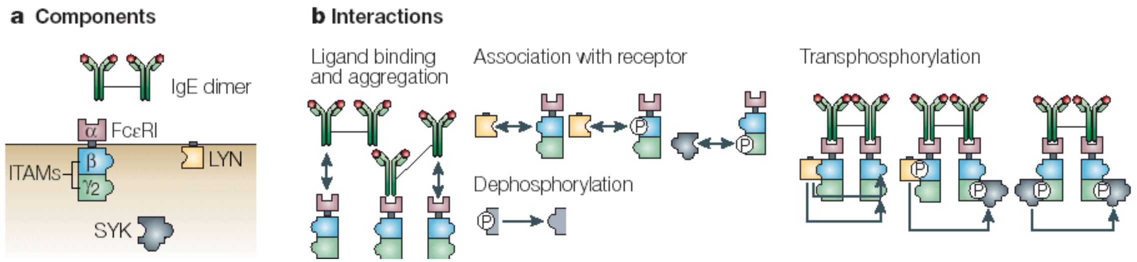
- *Concise and precise* representation of biochemical knowledge
  - rules are simple (less context) when interactions are modular
- *Flexible* with respect to simulation method
  - Deterministic / stochastic
  - Well-mixed / compartmental / spatial
- Structures and rules are *reusable*
  - Rule libraries
- Compact visual representation
  - Contact map and beyond



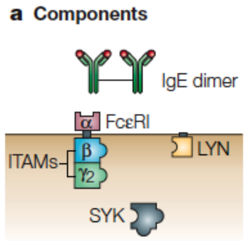

Contact map

# Rule-Based Modeling protocol

1. Identify components and interactions

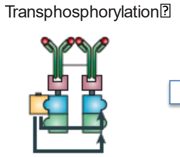



2. Translate into objects (molecules) and rules




```
IgE(a,a)
FceRI(a,b~U~P,g2~U~P)
Lyn(U,SH2)
Syk(tSH2,lY~U~P,aY~U~P)
```

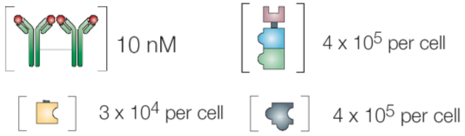
…




…

```
Lyn(U!1).FceRI(b!1).FceRI(b~U)-> \
Lyn(U!1).FceRI(b!1).FceRI(b~P)
```

3. Determine **concentrations** and **rate constants**




4. Simulate and analyze the model

ODE's

SSA

PDE's

BD

# MAPK model diagram